

THE FINITE-DEGREE PROFILE OF ESSENTIAL DIMENSION OF A $\mathbb{Z}/p^n$ -TORSOR IN CHARACTERISTIC p

ABSTRACT. For a G -torsor τ over a field $F \supseteq k$ and an integer $d \geq 1$ we study $\mathrm{ed}_k^{[d]}(\tau) = \min\{\mathrm{ed}_k(\tau \otimes_F F') : [F' : F] \leq d\}$, the smallest essential dimension that τ acquires over an extension of degree at most d , together with its jump degrees $d_j(\tau) = \min\{d : \mathrm{ed}_k^{[d]}(\tau) \leq j\}$. For $G = \mathbb{Z}/p^n$ in characteristic p and the generic torsor τ_{gen} over $K = k(s_0, \dots, s_{n-1})$ the profile is a non-increasing sequence starting at $\mathrm{ed}_k(\mathbb{Z}/p^n)$ and ending at 0 at $d = p^n$; it is invariant under purely inseparable base change; its prime-to- p tail is the essential dimension at p of Reichstein–Vistoli; and $\mathrm{ed}_k^{[p^j]}(\tau_{\mathrm{gen}}) \leq n - j$ by an explicit tower of j Artin–Schreier extensions. We give the geometric meaning of $\mathrm{ed}_k^{[d]}$ in terms of closed points of bounded degree in the free locus of twisted G -varieties, i.e. of G -equivariant correspondences of bounded degree in the sense of Reichstein–Scavia, prove that every “minimum over all G -varieties” invariant of the same kind collapses to a single bit, compute the profile for $n = 2$, and at $(p, n) = (2, 3)$ pin the open slots down to $d_1(\tau_{\mathrm{gen}}) \in \{2, 3, 4\}$ and $\mathrm{ed}^{[3]} \in \{1, 2\}$. Along the way we correct two statements of the companion note: the tower $K(v_1, v_2)$ with $v_i^2 + v_i = s_i$ does *not* make the generic $\mathbb{Z}/8$ -torsor one-dimensional (its essential dimension over that field is exactly 2), and the elliptic curve E'' used there has $j = 0$, so the argument must not rely on non-isotriviality; we give the correct argument.

1. INTRODUCTION

Let k be an algebraically closed field of characteristic $p > 0$ and $G = \mathbb{Z}/p^n$. Ledet’s conjecture [5] predicts $\mathrm{ed}_k(G) = n$; it is known for $n \leq 2$, and for $n \geq 3$ only $\mathrm{ed}_k(G) \in \{2, \dots, n\}$ is known. Two facts frame the problem. First, $\mathrm{ed}_k(G) = \mathrm{ed}_k(\tau_{\mathrm{gen}})$ for the generic torsor τ_{gen} over $K = k(s_0, \dots, s_{n-1})$ (Lemma 2.1). Second, over every *large* field $F \supseteq k$ (Laurent series fields, henselian fields, p -closed fields) every G -torsor has $\mathrm{ed}_k \leq 1$ [12, Lemma 9.1(b)], [13]. So every obstruction to $\mathrm{ed}_k(\tau_{\mathrm{gen}}) \leq n - 1$ is invisible after base change to any large field, and in particular after base change to the p -closure $K^{(p)}$, which is a union of finite extensions of degree prime to p .

This note studies what happens between K and its large extensions when the degree of the extension is controlled:

$$\mathrm{ed}_k^{[d]}(\tau) := \min\{\mathrm{ed}_k(\tau \otimes_F F') : F'/F \text{ a field extension, } [F' : F] \leq d\}.$$

The function $d \mapsto \mathrm{ed}_k^{[d]}(\tau_{\mathrm{gen}})$ is a non-increasing step function which starts at $\mathrm{ed}_k(G)$ and ends at 0; its steps are the jump degrees $d_j(\tau_{\mathrm{gen}})$. Its prime-to- p tail is the essential dimension at p in the sense of [12, (2.5)], and its value 1 is reached at some prime-to- p degree by the theorem of Reichstein–Vistoli. The p -power part is new: it records how many Artin–Schreier layers must be adjoined, and in which order, before the class becomes Teichmüller, and it is not seen by any large-field or cohomological invariant.

Results. Section 3 proves the basic properties (monotonicity, behaviour under composition of extensions, purely inseparable invariance, the truncation bound $\mathrm{ed}^{[p^j]}(\tau_{\mathrm{gen}}) \leq n - j$, the relation to $\mathrm{ed}_{k,p}$, and the value of ed over the intermediate fields of L/K). Section 4 gives the geometric meaning: $\mathrm{ed}_k^{[d]}(\tau) \leq m$ iff some generically free G -variety of dimension $\leq m$ has a closed point of degree $\leq d$ in the free locus of its twist by τ , i.e. iff a G -equivariant correspondence of degree $\leq d$ from the torsor into a free G -variety of dimension $\leq m$ exists (Theorem 4.2). Section 5 shows that the “global minimum” invariants (minimal degree of a free point, minimal genus of a free curve,

minimal transcendence degree) all collapse to one bit or to values $\leq p$, using a cheap non-versal hypersurface $S_0 \subset W_n$. Section 6 computes the profile at $n = 2$: it is $(2, 1, 1, 0)$ for $p = 2$, with the twisted elliptic curve E' : $w^2 + w = u(u+1)(u+s_0) + s_1$ over $k(s_0, s_1)$ having Mordell–Weil group 0; for $p \geq 3$ one has $2 \leq d_1(\tau_2) \leq p$. Section 7 treats $(p, n) = (2, 3)$:

$$\text{ed}^{[1]} \in \{2, 3\}, \quad \text{ed}^{[2]} \leq 2, \quad \text{ed}^{[3]} \in \{1, 2\}, \quad \text{ed}^{[4]} = 1, \quad \text{ed}^{[8]} = 0, \quad d_1(\tau_{\text{gen}}) \in \{2, 3, 4\},$$

and proves that over each of the three natural quadratic extensions $K(v_1)$, $K(z_0)$ and over the quartic tower $K(v_1, v_2)$ ($v_i^2 + v_i = s_i$, $z_0^2 + z_0 = s_0$) the essential dimension is exactly 2 (Theorem 7.3), so that a quadratic witness for $d_1 = 2$, if it exists, is of a different shape; the shape of a hypothetical witness on the Artin–Schreier–Witt curve is determined in Proposition 7.5. Section 8 records the per-surface free-degree set with $\delta_{\text{free}}(X_{0,2}) = 2$. Section 9 explains why the profile is a genuine intermediate invariant, lists the open questions and writes out the first computation as an algorithm.

Corrections to the companion note [4]. Two statements of [4, §2] are corrected here. (i) The field $K(v_{n-j}, \dots, v_{n-1})$ with $v_i^p - v_i = s_i$ is *not* in general a witness for $\text{ed}^{[p^j]}(\tau_{\text{gen}}) \leq n - j$: at $(p, n) = (2, 3)$, the class of τ_{gen} over $K(v_1, v_2)$ is $(s_0, 0, s_1^2 + s_1 v_1)$, which is not Teichmüller, and $\text{ed}_k(\tau_{\text{gen}} \otimes K(v_1, v_2)) = 2$ (Theorem 7.3). The bound itself is correct, with the witness replaced by a tower in which each new Artin–Schreier generator kills the carry produced by the previous one (Lemma 3.3). (ii) The elliptic curve E'' : $w^2 + w = u(u+1)(u+s_0)$ over $k(s_0)$ has j -invariant 0, hence is isotrivial; the statement $E''(F) = E''(k(s_0))$ for F purely transcendental over $k(s_0)$ used in [4] is nevertheless true, for every elliptic curve, by the elementary Lemma 4.1. Also, the class over $K(v_1)$ is $(s_0, 0, s_2 + s_1^2 + s_1 v_1)$ rather than $(s_0, 0, s_2)$; the argument survives (Remark 7.4), and is superseded by the stronger Theorem 7.3. Finally, $\text{ed}^{[d]}(\tau_{\text{gen}}) \geq 1$ holds for $d < p^n$ only: $\text{ed}^{[p^n]}(\tau_{\text{gen}}) = 0$, witnessed by $F' = L$.

Tags. Every numbered statement carries a tag: [PROVED] (complete proof here), [CITED] (quoted from the literature), [IMPORTED] (from the working notes [7, 15, 4], with the status stated there), [COMPUTED] (script in `tex_profile/scratch/`), [OPEN].

2. CONVENTIONS AND BACKGROUND

Throughout, k is an algebraically closed field of characteristic $p > 0$, all fields contain k , and $G = \mathbb{Z}/p^n = \langle g \rangle$ unless said otherwise.

2.1. Essential dimension. Let $\tau: T \rightarrow \text{Spec } F$ be a G -torsor over a field $F \supseteq k$. Following [2, 1, 9], $\text{ed}_k(\tau)$ is the least $\text{trdeg}_k F_0$ over subfields $k \subseteq F_0 \subseteq F$ such that $\tau \cong \tau_0 \otimes_{F_0} F$ for a G -torsor τ_0 over F_0 , and $\text{ed}_k(G) = \sup_{\tau} \text{ed}_k(\tau)$. For F'/F a field extension, $\text{ed}_k(\tau \otimes_F F') \leq \text{ed}_k(\tau)$ [12, Lemma 2.1(a)], and $\text{ed}_k(\tau) \geq \text{ed}_l(\tau)$ for $k \subseteq l \subseteq F$ [12, Lemma 2.1(b)]. Since k is algebraically closed, $\text{ed}_k(\tau) = 0$ iff τ is split.

Let Y be a generically free primitive G -variety over k (primitive: G permutes the irreducible components transitively). A G -compression is a dominant G -equivariant rational map $Y \dashrightarrow X$ to a generically free primitive G -variety, and $\text{ed}_k(Y)$ is the minimal $\dim X$ over compressions (for finite G). Shrinking Y to a G -torsor $Y^0 \rightarrow B$ with $k(B) = k(Y)^G =: F$ and pulling back to the generic point gives a G -torsor τ_Y over F , and

$$(1) \quad \text{ed}_k(Y) = \text{ed}_k(\tau_Y) \quad [12, (2.3)], [9, \text{Lemma 3.9}].$$

Essential dimension at a prime q : for a G -torsor τ over F , $\text{ed}_{k,q}(\tau)$ is the minimal value of $\text{ed}_k(\tau_{F'})$ over finite field extensions F'/F of degree prime to q ; equivalently $\text{ed}_{k,q}(\tau) = \text{ed}_k(\tau_{F^{(q)}})$ for a q -closure $F^{(q)}$ [12, (2.5)]. A correspondence $X \rightsquigarrow Z$ of degree d between G -varieties is a diagram $X \xleftarrow{\pi} X' \dashrightarrow Z$ of G -equivariant rational maps with π dominant of degree d [12, (2.6)].

2.2. Witt vectors and Artin–Schreier–Witt theory. For an $\mathbb{F}_p$ -algebra A , $W_n(A) = A^n$ with the ring structure defined over $\mathbb{Z}$ by the ghost components $w_m(x) = \sum_{i \leq m} p^i x_i^{p^{m-i}}$. We use the Frobenius $F(x) = (x_i^p)$, the Verschiebung $V(x) = (0, x_0, \dots, x_{n-2})$, the Teichmüller lift $[a] = (a, 0, \dots, 0)$ and $\wp = F - \text{id}$, an additive endomorphism. Standard facts: V is additive; $FV = VF = p$ on $W(A)$ when $pA = 0$, hence $\wp V = V\wp$; and $x + V(y) = (x_0, (x_1, \dots, x_{n-1}) + y)$, whence by iteration

$$(2) \quad (x_0, \dots, x_{r-1}, 0, \dots, 0) + V^r(y) = (x_0, \dots, x_{r-1}, y_0, \dots, y_{n-r-1}), \quad y \in W_{n-r}(A),$$

so that every $x = (x_0, \dots, x_{n-1})$ equals $(x_0, \dots, x_{r-1}, 0, \dots, 0) + V^r(x_r, \dots, x_{n-1})$. On $W_1(A) = A$ the Teichmüller map is the identity and $\wp(a) = a^p - a$. For $p = 2$, $n = 3$ the addition and $\wp$ read (script `witt_p2n3.py`)

$$(3) \quad (a+b)_1 = a_1 + b_1 + a_0 b_0, \quad (a+b)_2 = a_2 + b_2 + a_1 b_1 + a_0 b_0(a_1 + b_1) + a_0^3 b_0 + a_0 b_0^3,$$

$$(4) \quad \begin{aligned} \wp(a) &= (a_0^2 + a_0, a_1^2 + a_1 + a_0^3 + a_0^2, a_2^2 + a_2 + C(a_0, a_1)), \\ C(a_0, a_1) &= a_0^7 + a_0^4 + a_0^3 a_1^2 + a_0^3 a_1 + a_0^2 a_1^2 + a_0^2 a_1 + a_1^3 + a_1^2. \end{aligned}$$

Artin–Schreier–Witt theory [16]: for every field F of characteristic p ,

$$H^1(F, \mathbb{Z}/p^n) \cong W_n(F)/\wp W_n(F),$$

the class of a being the G -torsor $\tau_a = \text{Spec } F[x]/(\wp x = a)$ with $g \cdot x = x + 1$, $1 = (1, 0, \dots, 0)$. The truncation $W_n \rightarrow W_m$ is a ring map commuting with $\wp$ and corresponds to $\mathbb{Z}/p^n \rightarrow \mathbb{Z}/p^m$; the map $V^{n-m}: W_m \rightarrow W_n$ is additive, commutes with $\wp$ and corresponds to the inclusion $\mathbb{Z}/p^m \hookrightarrow \mathbb{Z}/p^n$ on cohomology. We shall use repeatedly:

$$(5) \quad V^{n-1}(c) = (0, \dots, 0, c) \in \wp W_n(F) \iff c \in \wp F.$$

(If $(0, \dots, 0, c) = \wp(x)$ then $x_0^p = x_0$, so $x = [x_0] + V(x')$ with $\wp[x_0] = 0$, hence $(0, \dots, 0, c) = V\wp(x')$; induct.) Also $a^{p^r} \equiv a \pmod{\wp F}$ for $a \in F$.

2.3. The generic torsor. G acts on $W_n = \mathbb{A}_k^n$ (coordinates $z = (z_0, \dots, z_{n-1})$) by the Witt translation $g(z) = z + 1$; $g^{p^j}(z) = z + V^j(1)$ fixes $z_0, \dots, z_{j-1}$ and acts on $(z_j, \dots, z_{n-1})$ as the Witt translation of W_{n-j} [8, Lemma 1.4]. The map $\wp: W_n \rightarrow W_n$ is finite étale Galois with group $\langle g \rangle$; writing $s = \wp(z)$, $L := k(z)$, $K := L^G = k(s_0, \dots, s_{n-1})$, the generic fibre $\tau_{\text{gen}}: \text{Spec } L \rightarrow \text{Spec } K$ is the G -torsor of class $s \in W_n(K)$, and L/K is a field extension of degree p^n whose intermediate fields are $K_j := L^{\langle g^{p^j} \rangle} = k(z_0, \dots, z_{j-1}, s_j, \dots, s_{n-1})$, $[K_j : K] = p^j$. When several n are in play we write τ_n , $K_{(n)}$, $L_{(n)}$. The action of $g^{p^{n-1}}$, $z \mapsto (z_0, \dots, z_{n-2}, z_{n-1} + 1)$, has no fixed points, so G acts freely on W_n .

Lemma 2.1 ([15], Lemma A.4). W_n is τ -versal for every G -torsor τ over every field $F \supseteq k$, and $\text{ed}_k(G) = \text{ed}_k(W_n) = \text{ed}_k(\tau_{\text{gen}}) \leq n$. [IMPORTED]

2.4. Twists, free locus, versality, large fields. For a G -variety X over k and a G -torsor $\tau: T \rightarrow \text{Spec } F$, the twist is ${}^\tau X = (T \times_k X)/G$, an algebraic space over F (a variety if X is quasi-projective) [3]. It commutes with base change in F , and for a G -stable open $U \subseteq X$, ${}^\tau U \subseteq {}^\tau X$ is an open subspace [3, Cor. 3.4]. We quote from [12]:

Fact 2.2 ([12], §7, Lemmas 7.3, 7.4, 9.1). *Let X be a G -variety, $\tau: T \rightarrow \text{Spec } F$ a G -torsor. X is τ -weakly versal if there is a G -equivariant morphism $T \rightarrow X_F$ over F , and τ -versal if every dense G -invariant open subvariety is τ -weakly versal.*

- (a) X is τ -weakly versal iff ${}^\tau X(F) \neq \emptyset$. More precisely ${}^\tau X(F) = \text{Hom}_G(T, X_F) = \text{Hom}_G^k(T, X)$.
- (b) If X is smooth and F is large, X is weakly τ -versal iff X is τ -versal.
- (c) If X is generically free and primitive and τ -versal, then $\text{ed}_k(X) \geq \text{ed}_k(\tau)$.

- (d) If F is large and X generically free with a smooth G -fixed k -point, then $\text{ed}_k(\tau) \leq \text{ed}_k(X) \leq \dim X$. If G is a finite p -group, $\text{ed}_k(\tau) \leq 1$ for every torsor over a large field.

A field F is large if every irreducible F -variety with a smooth F -point has dense F -points; $F((t))$, henselian nontrivially valued fields, and q -closed fields are large [12, §7], [10, 11]. [CITED]

For a G -variety X put $X^{\text{free}} := X \setminus \bigcup_{1 \neq h \in G} X^h$, a G -stable open subset. Since every nontrivial subgroup of the cyclic p -group G contains $\langle g^{p^{n-1}} \rangle$, $X^{\text{free}} = X \setminus X^{g^{p^{n-1}}}$. The free locus of the twist is ${}^\tau(X^{\text{free}}) \subseteq {}^\tau X$; an F -point of ${}^\tau X$ is free if it lies there, i.e. (by Fact 2.2(a) applied to X^{free}) if the corresponding equivariant morphism $T \rightarrow X$ factors through X^{free} . A closed point x of ${}^\tau X$ has degree $[\kappa(x) : F]$.

Lemma 2.3. *Let Z be a quasi-projective generically free primitive G -variety over k and τ a G -torsor over F . If ${}^\tau Z$ has a free F -point, then $\text{ed}_k(\tau) \leq \dim Z$.* [PROVED]

Proof. G acts freely on the quasi-projective variety Z^{free} , so the quotient $\pi : Z^{\text{free}} \rightarrow B := Z^{\text{free}}/G$ exists and is a G -torsor (SGA 1, V). A free F -point is a G -equivariant $f : T \rightarrow Z^{\text{free}}$; it induces $\bar{f} : \text{Spec } F \rightarrow B$ with $f^*(\pi) \cong \tau$ (a G -equivariant map of torsors over $\text{Spec } F$ is an isomorphism). If b is the image point of $\bar{f}$, then τ is the base change of the torsor $\pi^{-1}(b) \rightarrow \text{Spec } \kappa(b)$ along $\kappa(b) \subseteq F$, and $\text{trdeg}_k \kappa(b) \leq \dim B = \dim Z$. $\square$

Lemma 2.4 (image of a free point). *Let Z be an irreducible G -variety with faithful action, $\tau : T \rightarrow \text{Spec } F$ a torsor with T connected, and $f : T \rightarrow Z$ equivariant. Then the closure of $f(T)$ is an irreducible G -stable closed subset Y , and the point $f(T)$ is free iff G acts faithfully on Y . If $\dim Z = 1$ and $f(T)$ is free, f is dominant.* [PROVED]

Proof. As T is connected and Z separated, $\overline{f(T)}$ is a single point ξ (the image of the generic point of T), and $g\xi = \xi$ by equivariance, so $Y = \{\xi\}$ is G -stable and irreducible. For $h \neq 1$, $\xi \in Z^h$ iff $Y \subseteq Z^h$ iff h acts trivially on Y . As all nontrivial subgroups contain $g^{p^{n-1}}$, ξ is free iff $g^{p^{n-1}}$ is nontrivial on Y iff the action on Y is faithful. A G -stable closed point is fixed by G , hence not free; so for a curve a free point has $Y = Z$. $\square$

Lemma 2.5 (curves). (i) An irreducible curve dominated by $\mathbb{A}_k^m$ has smooth projective model $\mathbb{P}^1$. (ii) $\text{PGL}_2(k)$ has no element of order p^2 . (iii) Let $p = 2$ and E a supersingular elliptic curve over k . Then $\text{Aut}(E)$ (all automorphisms of the curve) has no element of order 8. (iv) A nonconstant morphism of smooth projective curves $E \rightarrow C$ satisfies $g(C) \leq g(E)$. [PROVED]

Proof. (i) is Lüroth's theorem, valid in every characteristic (restrict the dominant map to a general line to get a nonconstant $\mathbb{P}^1 \rightarrow \tilde{C}$). (ii) A p -element of $\text{PGL}_2(k)$ lifts to $A \in \text{GL}_2(k)$ with $A^{p^m} = 1$ after scaling, so $(A - 1)^{p^m} = 0$, $A = 1 + N$ with $N^2 = 0$, and $A^p = 1 + pN = 1$. (iii) $\text{Aut}(E) = E(k) \rtimes \text{Aut}(E, O)$, and $\text{Aut}(E, O)$ is a group of order 24 with Sylow 2-subgroup Q_8 [14, App. A, Prop. 1.2], so its elements of 2-power order have order ≤ 4 . Let $\phi = t_P \circ \alpha$ with $\alpha \in \text{Aut}(E, O)$, t_P translation by P , and $\phi^8 = 1$; then $\alpha^8 = 1$, so $m := \text{ord } \alpha \in \{1, 2, 4\}$. If $m = 1$, $\text{ord } \phi = \text{ord } P$, which is odd because $E(k)[2] = 0$ for a supersingular curve in characteristic 2. If $m \in \{2, 4\}$: $\phi^m = t_Q$ with $Q = (1 + \alpha + \cdots + \alpha^{m-1})P$, and $(\alpha - 1)(1 + \alpha + \cdots + \alpha^{m-1}) = \alpha^m - 1 = 0$ in the domain $\text{End}(E)$ with $\alpha - 1 \neq 0$, so $Q = 0$ and $\text{ord } \phi \mid m \leq 4$. (iv) Factor $E \rightarrow C$ as a purely inseparable map (a power of Frobenius composed with an isomorphism, genus preserving) followed by a separable map, to which Riemann–Hurwitz applies. $\square$

3. THE PROFILE AND ITS BASIC PROPERTIES

Definition 3.1. Let τ be a G -torsor over a field $F \supseteq k$ and $d \geq 1$. Put

$$\text{ed}_k^{[d]}(\tau) := \min\{\text{ed}_k(\tau \otimes_F F') : [F' : F] \leq d\}, \quad d_j(\tau) := \min\{d \geq 1 : \text{ed}_k^{[d]}(\tau) \leq j\} \quad (0 \leq j \leq n).$$

The sequence $(\mathrm{ed}_k^{[d]}(\tau))_{d \geq 1}$ is the *finite-degree profile* of τ and the $d_j(\tau)$ are its *jump degrees*. We also write $d_j^{(p')}(\tau) := \min\{d \text{ prime to } p : \mathrm{ed}_k^{[d]}(\tau) \leq j\}$.

Proposition 3.2 (shape). *Let τ be a G -torsor over $F \supseteq k$.*

- (a) $d \mapsto \mathrm{ed}_k^{[d]}(\tau)$ is non-increasing with values in $\{0, \dots, \mathrm{ed}_k(\tau)\}$, $\mathrm{ed}_k^{[1]}(\tau) = \mathrm{ed}_k(\tau)$, and $d_{\mathrm{ed}(\tau)} = 1 \leq d_{\mathrm{ed}(\tau)-1} \leq \dots \leq d_0$.
- (b) If F'/F is a field extension of degree d and $d' \geq 1$, then $\mathrm{ed}_k^{[dd']}(\tau) \leq \mathrm{ed}_k^{[d']}(\tau \otimes F') \leq \mathrm{ed}_k^{[d]}(\tau)$.
- (c) $\mathrm{ed}_k^{[d]}(\tau) = 0$ iff τ is split by an extension of degree $\leq d$. For $\tau = \tau_{\mathrm{gen}}: \mathrm{ed}_k^{[d]}(\tau_{\mathrm{gen}}) \geq 1$ for $d < p^n$ and $\mathrm{ed}_k^{[p^n]}(\tau_{\mathrm{gen}}) = 0$, i.e. $d_0(\tau_{\mathrm{gen}}) = p^n$.

[PROVED]

Proof. (a) is clear from the definition and $\mathrm{ed}_k(\tau \otimes F') \leq \mathrm{ed}_k(\tau)$. (b) The first inequality: an extension F''/F' of degree $\leq d'$ has $[F'' : F] \leq dd'$. The second: if $[F_1 : F] \leq d'$ then the compositum $F_1 F'$ (any composite field) has $[F_1 F' : F'] \leq d'$ and $\mathrm{ed}_k(\tau \otimes F_1 F') \leq \mathrm{ed}_k(\tau \otimes F_1)$. (c) $\mathrm{ed}_k = 0$ iff split (k algebraically closed). $\tau_{\mathrm{gen}} \otimes F'$ is split iff $L \otimes_K F' \cong F'^{p^n}$ iff there is a K -embedding $L \hookrightarrow F'$, which forces $[F' : K] \geq p^n$; and $F' = L$ works. $\square$

Lemma 3.3 (truncation tower). *Let F be a field of characteristic p , $a \in W_n(F)$ and $0 \leq r \leq n$. There is a tower $F = F^{(0)} \subseteq F^{(1)} \subseteq \dots \subseteq F^{(n-r)}$ of Artin–Schreier extensions (each of degree 1 or p), hence $[F^{(n-r)} : F] \leq p^{n-r}$, such that*

$$a \equiv (a_0, \dots, a_{r-1}, 0, \dots, 0) \bmod \wp W_n(F^{(n-r)}).$$

The i -th layer is $F^{(i)} = F^{(i-1)}(v_i)$ with $v_i^p - v_i = b_i$, where $b_1 = a_r$ and b_i (for $i \geq 2$) is the coordinate of index $r+i-1$ of the class obtained at the previous step; in general $b_i \neq a_{r+i-1}$ because of the Witt carries.

[PROVED]

Proof. Induction on $n-r$; nothing to do if $r = n$. Let $r < n$. By (2), $a = (a_0, \dots, a_{r-1}, 0, \dots, 0) + V^r(a')$ with $a' = (a_r, \dots, a_{n-1}) \in W_{n-r}(F)$, and $a' = [a_r] + V(a'')$. Let $v^p - v = a_r$ in $F^{(1)} = F(v)$. In $W_{n-r}(F^{(1)})$, $[a_r] = [v^p - v]$ and $\wp[v] = [v^p] - [v]$ have the same 0-th coordinate, so $[a_r] = \wp[v] + V(e)$ for some $e \in W_{n-r-1}(F^{(1)})$. Since $\wp$ commutes with V^r , $V^r \wp[v] \in \wp W_n(F^{(1)})$ and

$$a \equiv (a_0, \dots, a_{r-1}, 0, \dots, 0) + V^{r+1}(a'' + e) = (a_0, \dots, a_{r-1}, 0, b_2, \dots, b_{n-r}) \bmod \wp W_n(F^{(1)}),$$

with $b_2, \dots$ the coordinates of $a'' + e$ (again (2)). Apply the induction hypothesis (with $r+1$) to this vector over $F^{(1)}$. $\square$

Proposition 3.4 (truncation bound). *For $0 \leq j \leq n$: $\mathrm{ed}_k^{[p^j]}(\tau_{\mathrm{gen}}) \leq n - j$. In particular $d_{n-1}(\tau_{\mathrm{gen}}) \leq p$, $d_1(\tau_{\mathrm{gen}}) \leq p^{n-1}$, $d_0(\tau_{\mathrm{gen}}) \leq p^n$. More generally, for any G -torsor τ_a over F one has $\mathrm{ed}_k^{[p^j]}(\tau_a) \leq \mathrm{ed}_k(\tau_{(a_0, \dots, a_{n-j-1}, 0, \dots, 0)}) \leq \mathrm{trdeg}_k k(a_0, \dots, a_{n-j-1})$.*

[PROVED]

Proof. Lemma 3.3 with $r = n - j$: over $F^{(j)}$ of degree $\leq p^j$ the class is $(a_0, \dots, a_{n-j-1}, 0, \dots, 0)$, a Witt vector with coordinates in $k(a_0, \dots, a_{n-j-1})$, so the torsor descends to that field. For τ_{gen} , $\mathrm{trdeg}_k k(s_0, \dots, s_{n-j-1}) = n - j$. $\square$

Remark 3.5. For $n = 2$ no carry occurs: $s = [s_0] + V(s_1)$ and $V(s_1) = V(\wp v) = \wp V(v)$ over $F(v)$, $v^p - v = s_1$, so $s \equiv [s_0]$ over $K(v)$. For $(p, n) = (2, 3)$ the script `witt_p2n3.py` gives, with $v_i^2 + v_i = s_i$,

$$s \equiv (s_0, 0, c) \text{ over } K(v_1), \quad c = s_2 + s_1^2 + s_1 v_1; \quad s \equiv (s_0, 0, s_1^2 + s_1 v_1) \text{ over } K(v_1, v_2);$$

$$s \equiv (s_0, 0, 0) \text{ over } K(v_1, v'),$$

where $v'^2 + v' = c$. The middle class is not Teichmüller: $s_1^2 + s_1 v_1 \equiv v_1^3 + v_1 \bmod \wp K(v_1, v_2)$, and $v_1^3 + v_1 \notin \wp k(s_0, v_2)(v_1)$ since it has a pole of odd order 3 at $v_1 = \infty$ while $w^2 + w$ has even or

non-negative valuation there. So $K(v_1, v_2)$ is not a witness for $\text{ed}^{[4]}(\tau_{\text{gen}}) \leq 1$; the tower $K(v_1, v')$ is (in fact $\text{ed}_k(\tau_{\text{gen}} \otimes K(v_1, v_2)) = 2$, Theorem 7.3). This corrects [4, Prop. 2.2(c) and §2.6]. [COMPUTED]

Proposition 3.6 (purely inseparable invariance). *Let τ be a G -torsor over $F \supseteq k$ and F'/F finite purely inseparable. Then $\text{ed}_k(\tau \otimes_F F') = \text{ed}_k(\tau)$. Hence the profile only sees separable extensions: $\text{ed}_k^{[d]}(\tau) = \min\{\text{ed}_k(\tau \otimes F') : [F' : F] \leq d, F'/F \text{ separable}\}$.* [PROVED]

Proof. Choose m with $F' \subseteq F^{1/p^m}$ (inside an algebraic closure). Then $\text{ed}_k(\tau \otimes F^{1/p^m}) \leq \text{ed}_k(\tau \otimes F') \leq \text{ed}_k(\tau)$, and it suffices to prove $\text{ed}_k(\tau \otimes F^{1/p^m}) \geq \text{ed}_k(\tau)$. Let $\sigma : F^{1/p^m} \rightarrow F$, $x \mapsto x^{p^m}$; it is a field isomorphism with $\sigma(k) = k$ since k is perfect. For any G -torsor τ' over F^{1/p^m} let $\sigma_*\tau'$ be the torsor over F obtained by transport of structure. The map $F_0 \mapsto \sigma(F_0)$ is a bijection between subfields of F^{1/p^m} containing k and subfields of F containing k , preserving trdeg_k , and τ' descends to F_0 iff $\sigma_*\tau'$ descends to $\sigma(F_0)$; so $\text{ed}_k(\sigma_*\tau') = \text{ed}_k(\tau')$. Now let $a \in W_n(F)$ be the class of τ ; the class of $\tau \otimes F^{1/p^m}$ is a viewed in $W_n(F^{1/p^m})$, and by functoriality of the Artin–Schreier–Witt isomorphism the class of $\sigma_*(\tau \otimes F^{1/p^m})$ is $W_n(\sigma)(a) = F^m(a)$. Since $F(a) - a = \wp(a)$, $F^m(a) \equiv a \pmod{\wp W_n(F)}$, so $\sigma_*(\tau \otimes F^{1/p^m}) \cong \tau$ and $\text{ed}_k(\tau \otimes F^{1/p^m}) = \text{ed}_k(\tau)$. The last statement: any finite F'/F is purely inseparable over its separable closure F'_s in F' , $[F'_s : F] \leq [F' : F]$, and $\text{ed}_k(\tau \otimes F') = \text{ed}_k(\tau \otimes F'_s)$. $\square$

Proposition 3.7 (the prime-to- p tail). *Let τ be a G -torsor over $F \supseteq k$, G a finite p -group. Put $\text{ed}_k^{[d]'}(\tau) := \min\{\text{ed}_k(\tau \otimes F') : [F' : F] \leq d, p \nmid [F' : F]\}$, the prime-to- p profile, so that $\text{ed}_k^{[d]} \leq \text{ed}_k^{[d]'}$.*

- (a) $\text{ed}_{k,p}(\tau) = \min_d \text{ed}_k^{[d]' }(\tau)$, and $\text{ed}_k(\tau \otimes F') \geq \text{ed}_{k,p}(\tau)$ for every F'/F of degree prime to p .
Thus $\text{ed}_{k,p}$ is the terminal value of the prime-to- p profile, and $\min_{p \nmid d} \text{ed}_k^{[d]}(\tau) \leq \text{ed}_{k,p}(\tau)$.
- (b) $\text{ed}_{k,p}(\tau) \leq 1$; hence $\text{ed}_k^{[d]' }(\tau) \leq 1$ for some d , and for $\tau = \tau_{\text{gen}}$ there is a prime-to- p degree d with $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k^{[d]' }(\tau_{\text{gen}}) = 1$, while $\text{ed}_k^{[d]}(\tau_{\text{gen}}) \geq 1$ for all $d < p^n$; thus $d_1^{(p') }(\tau_{\text{gen}}) < \infty$.
- (c) For $G = \mathbb{Z}/p^n$ the witness in (b) can be taken on the Artin–Schreier–Witt curve: there are a finite prime-to- p extension F'/F and $t \in F'$ with $\tau \otimes F' \cong \tau_{[t]}$ (Teichmüller form).

[PROVED]

Proof. (a) Put $e := \text{ed}_k(\tau \otimes F^{(p)}) = \text{ed}_{k,p}(\tau)$ [12, (2.5)]. A descent of $\tau \otimes F^{(p)}$ to a subfield of transcendence degree e involves finitely many elements of $F^{(p)}$, all lying in a finite subextension $F_1 \subset F^{(p)}$, of degree prime to p ; so $\text{ed}_k(\tau \otimes F_1) \leq e$ and $\text{ed}_k^{[d]' }(\tau) \leq e$ for $d = [F_1 : F]$. Conversely, for any F'/F with $p \nmid [F' : F]$, $\text{ed}_k(\tau \otimes F') \geq \text{ed}_k(\tau \otimes F'F^{(p)})$, and $F'F^{(p)}$ is algebraic over $F^{(p)}$ of degree prime to p , hence equal to $F^{(p)}$ (p -closed); so $\text{ed}_k(\tau \otimes F') \geq e$.

(b) The first claim is Fact 2.2(d) over the large field $F^{(p)}$ together with [12, (2.5)]. By (a) there is a prime-to- p F_1 with $\text{ed}_k(\tau \otimes F_1) \leq 1$; for τ_{gen} , $\text{ed}^{[d]} \geq 1$ for $d < p^n$ by Proposition 3.2(c), so $\text{ed}^{[d]}(\tau_{\text{gen}}) = \text{ed}^{[d]' }(\tau_{\text{gen}}) = 1$ for $d = [F_1 : K]$.

(c) Let $\bar{U}_n$ be the smooth projective model of the Artin–Schreier–Witt curve $U_n = \{y \in W_n : \wp(y) \in [\mathbb{A}^1]\}$; $U_n \rightarrow \mathbb{A}^1$, $y \mapsto t = y_0^p - y_0$, is a connected G -Galois cover (its generic fibre is the torsor of class $[t] \in W_n(k(t))$, of order p^n : $p^{n-1}[t] = V^{n-1}F^{n-1}[t] = (0, \dots, 0, t^{p^{n-1}})$ is $\neq 0$ by (5) since $t^{p^{n-1}} \equiv t \notin \wp k(t)$). It is étale over $\mathbb{A}^1$ and totally ramified over ∞ (the quotient by the inertia group is a connected étale cover of $\mathbb{P}^1$, hence trivial), so ∞ is a smooth G -fixed k -point of $\bar{U}_n$, and G acts freely on U_n (the affine part of $\bar{U}_n^{\text{free}}$ is U_n). The constant map gives a smooth F -point $\infty' \in \bar{U}_n(F)$. Over the large field $F^{(p)}$ the points are dense, so one lies in the dense open $\mathcal{T}U_n$; it is defined over a finite prime-to- p subextension F' . A free F' -point is an equivariant $T_{F'} \rightarrow U_n$, i.e. $y \in W_n(L \otimes_F F')$ with $gy = y + 1$ and $\wp(y) = [t]$, $t \in F'$: then $y - x$ is G -invariant for any x with $\wp x = a$ (the class of τ), so $y = x + h$ with $h \in W_n(F')$ and $a + \wp h = [t]$. $\square$

Remark 3.8. $\text{ed}^{[d]}$ for $p \nmid d$ minimises over all extensions of degree $\leq d$, including some of degree divisible by p , which is why (a) is stated with the prime-to- p profile. For τ_{gen} the number $d_1^{(p')}(\tau_{\text{gen}})$ (the prime-to- p jump degree) is the honest refinement of the statement $\text{ed}_{k,p}(\mathbb{Z}/p^n) = 1$ of [13]: the latter says only that $d_1^{(p')} < \infty$.

Proposition 3.9 (the intermediate fields of L/K). *Let $K_j = L^{\langle g^{p^j} \rangle}$, $1 \leq j \leq n-1$. Then $\tau_{\text{gen}} \otimes K_j$ is induced from the $\mathbb{Z}/p^{n-j}$ -torsor L/K_j , and*

$$\text{ed}_k(\mathbb{Z}/p^{n-j}) \leq \text{ed}_k(\tau_{\text{gen}} \otimes K_j) \leq n-j.$$

In particular $\text{ed}_k(\tau_{\text{gen}} \otimes K_1) \geq \min(n-1, 2)$, so for $n=3$ the subextension $K_1 = K(z_0)$ is never a witness for $d_1 = p$. [PROVED]

Proof. $L \otimes_K K_j \cong \prod_{G/H} L$ with $H = \langle g^{p^j} \rangle$, i.e. $\tau_{\text{gen}} \otimes K_j = \text{Ind}_H^G(L/K_j)$. By [8, Lemma 1.4], H acts on $(z_j, \dots, z_{n-1})$ by the Witt translation of W_{n-j} , so L/K_j is the $\mathbb{Z}/p^{n-j}$ -torsor of class $s' := \wp_{W_{n-j}}(z_j, \dots, z_{n-1}) \in W_{n-j}(K_j)$. Its coordinates satisfy $s'_i = z_{j+i}^p - z_{j+i} + (\text{polynomial in } z_j, \dots, z_{j+i-1})$, while $s_{j+i} = z_{j+i}^p - z_{j+i} + (\text{polynomial in } z_0, \dots, z_{j+i-1})$; hence $s_{j+i} - s'_i$ is a polynomial in $z_0, \dots, z_{j+i-1}$ that is H -invariant, so it lies in $k(z_0, \dots, z_{j-1})(s'_0, \dots, s'_{i-1})$ by induction on i . Consequently

$$K_j = k'(s'_0, \dots, s'_{n-j-1}), \quad k' := k(z_0, \dots, z_{j-1}),$$

with s' algebraically independent over k' : L/K_j is the generic $\mathbb{Z}/p^{n-j}$ -torsor over the base field k' . Lemma 2.1 over k' gives $\text{ed}_{k'}(L/K_j) = \text{ed}_{k'}(\mathbb{Z}/p^{n-j}) \geq \text{ed}_{\bar{k}'}(\mathbb{Z}/p^{n-j}) = \text{ed}_k(\mathbb{Z}/p^{n-j})$ [12, Lemma 2.4(c),(d)], and $\text{ed}_k \geq \text{ed}_{k'}$ [12, Lemma 2.1(b)]. It remains to see $\text{ed}_k(\text{Ind}_H^G \tau_H) \geq \text{ed}_k(\tau_H)$ for an H -torsor τ_H over F whose total space is a field (the reverse inequality is clear: descend τ_H , then induce). Let $T_0 \rightarrow \text{Spec } F_0$ be a descent of $\text{Ind}_H^G \tau_H$, $F_0 \subseteq F$. Let T'_0 be a connected component of T_0 , with stabiliser H_0 ; T'_0 is an H_0 -torsor over F_0 with T'_0 a field, and $T_0 \otimes_{F_0} F = \coprod_{G/H_0} T'_0 \otimes_{F_0} F$. Comparing with $\text{Ind}_H^G \tau_H$ (connected components are H -torsors that are fields, and G is abelian) we get $H \subseteq H_0$ and $T'_0 \otimes_{F_0} F \cong \text{Ind}_H^{H_0} T_H$. Let $F_1 := (T'_0)^H$, a finite extension of F_0 inside T'_0 ; then T'_0 is an H -torsor over F_1 with T'_0 a field, and

$$T'_0 \otimes_{F_0} F = T'_0 \otimes_{F_1} (F_1 \otimes_{F_0} F) = \prod_i T'_0 \otimes_{F_1} F_i$$

over the fields F_i that are the factors of $F_1 \otimes_{F_0} F$. Each $T'_0 \otimes_{F_1} F_i$ is an H -torsor over F_i and a union of connected components of $T'_0 \otimes_{F_0} F$; these components are H -stable (H is their stabiliser) and H acts transitively on the geometric fibre, so $T'_0 \otimes_{F_1} F_i$ is connected and is one of the components, isomorphic to T_H as a scheme with H -action. Taking H -invariants, $F_i \cong T_H^H = F$ over F ; so F_1 embeds in F over F_0 and $\tau_H \cong T'_0 \otimes_{F_1} F$ descends to F_1 , with $\text{trdeg}_k F_1 = \text{trdeg}_k F_0$. $\square$

4. GEOMETRIC INTERPRETATION

Lemma 4.1 (purely transcendental base change). (i) *Let G be a finite group, τ a G -torsor over an infinite field $F \supseteq k$, and t transcendental over F . Then $\text{ed}_k(\tau \otimes_F F(t)) = \text{ed}_k(\tau)$.* (ii) *Let E be an elliptic curve (or any abelian variety) over a field F and $F' = F(t_1, \dots, t_m)$ purely transcendental. Then $E(F') = E(F)$.* [PROVED]

Proof. (i) $\leq$ is [12, Lemma 2.1(a)]. Conversely let $\tau \otimes F(t) \cong \tau_0 \otimes_{F_0} F(t)$ with $F_0 \subseteq F(t)$, $\text{trdeg}_k F_0 = e$; replacing F_0 by the subfield generated over k by the finitely many coefficients of equations defining τ_0 , we may assume F_0 finitely generated. Spread τ_0 out to a G -torsor $Z \rightarrow B_0$ over an integral k -variety B_0 with $k(B_0) = F_0$, $\dim B_0 = e$. The inclusion $F_0 \subseteq F(t)$ is a dominant rational map $\mathbb{A}_F^1 \dashrightarrow B_0$, defined on a dense open U , and the isomorphism of torsors over $F(t)$ spreads out to an isomorphism between the pullback of Z and the constant torsor $\tau \times U'$ over a dense open $U' \subseteq U$. Since F is infinite, U' has an F -point c ; restricting to c , τ is the pullback of Z along $\text{Spec } F \rightarrow B_0$, whose image point b has $\text{trdeg}_k \kappa(b) \leq e$. Hence $\text{ed}_k(\tau) \leq e$. (ii) By induction

it suffices to treat $m = 1$. A point of $E(F(t))$ is a rational map $\mathbb{P}_F^1 \dashrightarrow E$, which extends to a morphism $\phi: \mathbb{P}_F^1 \rightarrow E$ since E is proper and $\mathbb{P}_F^1$ is a regular curve. After base change to $\bar{F}$, ϕ is constant (a nonconstant morphism $\mathbb{P}^1 \rightarrow E$ would make $E_{\bar{F}}$ unirational, hence rational by Lüroth, contradicting $g = 1$; or use Lemma 2.5(iv)). So $\phi(\mathbb{P}_F^1)$ is a single closed point, which is the image of the F -point $\infty \in \mathbb{P}^1(F)$, hence is an F -point of E . $\square$

Theorem 4.2 (geometric meaning of the profile). *Let $\tau: T \rightarrow \operatorname{Spec} F$ be a G -torsor, $F \supseteq k$, and $m, d \geq 0$. The following are equivalent:*

- (1) $\operatorname{ed}_k^{[d]}(\tau) \leq m$;
- (2) *there is a quasi-projective generically free primitive G -variety Z over k with $\dim Z \leq m$ such that ${}^\tau Z$ has a closed point of degree $\leq d$ in the free locus ${}^\tau(Z^{\text{free}})$;*
- (3) *there are Z as in (2) and a field F'/F with $[F' : F] \leq d$ such that ${}^{\tau \otimes F'} Z$ has a free F' -point, i.e. a G -equivariant morphism $T \times_F F' \rightarrow Z^{\text{free}}$.*

For $\tau = \tau_{\text{gen}}$, (3) says: there is a G -equivariant rational map $W' \dashrightarrow Z^{\text{free}}$ from the normalisation W' of W_n in $L \otimes_K F'$, a (possibly reducible) G -variety dominating W_n with degree $[F' : K] \leq d$; when $L \otimes_K F'$ is a field this is a G -equivariant correspondence $W_n \rightsquigarrow Z$ of degree $\leq d$ in the sense of [12, (2.6)] landing in Z^{free} , and Z can then be taken irreducible with faithful G -action. [PROVED]

Proof. (2) $\Leftrightarrow$ (3): a closed point x of ${}^\tau Z$ with residue field $F' = \kappa(x)$ is an F' -point of ${}^\tau Z \otimes_F F' = {}^{\tau \otimes F'} Z$, free iff x lies in the open free locus (which commutes with base change); conversely the image of $\operatorname{Spec} F' \rightarrow {}^{\tau \otimes F'} Z \rightarrow {}^\tau Z$ is a point with residue field contained in F' , hence finite over F , hence closed, of degree $\leq [F' : F]$, and free if the F' -point is. The identification of free F' -points with equivariant maps $T_{F'} \rightarrow Z^{\text{free}}$ is Fact 2.2(a).

(3) $\Rightarrow$ (1): Lemma 2.3 gives $\operatorname{ed}_k(\tau \otimes F') \leq \dim Z \leq m$.

(1) $\Rightarrow$ (3): let $[F' : F] \leq d$ and $\tau \otimes F' \cong \tau_0 \otimes_{F_0} F'$ with $\operatorname{trdeg}_k F_0 \leq m$. Spread τ_0 out to a G -torsor $Z \rightarrow B$ over an integral affine k -variety B with $k(B) = F_0$, $\dim B \leq m$. Then Z is quasi-projective, G acts freely (so $Z^{\text{free}} = Z$) and Z is primitive (its generic fibre T_0 is a torsor over the field F_0 , on whose components G acts transitively). The composite $T_{F'} \rightarrow T_0 \rightarrow Z$ is G -equivariant.

For τ_{gen} : F' is a finite extension of $K = k(\mathbb{A}^n)$, so it is the function field of a variety B' with a finite dominant map $B' \rightarrow \mathbb{A}^n = W_n/G$ of degree $[F' : K]$, and $W' :=$ normalisation of $W_n \times_{\mathbb{A}^n} B'$ is a G -variety (action on the first factor) with $k(W') = L \otimes_K F'$ (a product of fields) and $W' \rightarrow W_n$ dominant of degree $[F' : K]$. An equivariant $\operatorname{Spec}(L \otimes_K F') \rightarrow Z^{\text{free}}$ is an equivariant rational map $W' \dashrightarrow Z^{\text{free}}$. If $L \otimes_K F'$ is a field, $T_{F'}$ is connected, hence so is T_0 , so τ_0 is a field and Z is irreducible with faithful action. $\square$

Definition 4.3. For a quasi-projective generically free primitive G -variety Z and a torsor τ over F , let $D(Z, \tau) \subseteq \mathbb{Z}_{>0}$ be the set of degrees of closed points of ${}^\tau Z$ lying in the free locus, and $\delta_{\text{free}}(Z, \tau) := \min D(Z, \tau)$ ($= \infty$ if empty). Write $D(Z)$, $\delta_{\text{free}}(Z)$ for $\tau = \tau_{\text{gen}}$.

Corollary 4.4. *For every G -torsor τ and $0 \leq j \leq n$,*

$$d_j(\tau) = \min\{\delta_{\text{free}}(Z, \tau) : Z \text{ quasi-projective generically free primitive, } \dim Z \leq j\}.$$

For $\tau = \tau_{\text{gen}}$ and $j = 1$ the minimum may be restricted to smooth projective irreducible curves with faithful G -action provided $d_1(\tau_{\text{gen}})$ is attained by an F' with $L \otimes_K F'$ a field (e.g. whenever $F' \not\supseteq K_1 = K(z_0)$, in particular whenever $p \nmid [F' : K]$); and then the free point is dominant (Lemma 2.4). [PROVED]

Proof. The first statement is Theorem 4.2. For the second: Z is an irreducible faithful G -curve, its smooth projective model $\tilde{Z}$ carries the induced action, the free F' -point of ${}^\tau Z$ is dominant (Lemma 2.4) hence corresponds to a G -equivariant $T_{F'} \rightarrow \tilde{Z}$ whose image is the generic point, free because the action on $\tilde{Z}$ is faithful; conversely a free point on $\tilde{Z}$ is dominant, so lands in the open Z .

The intermediate fields of L/K are the chain $K \subset K_1 \subset \cdots \subset L$, so a finite F'/K has $F' \cap L = K$ (i.e. $L \otimes_K F'$ a field, L/K being Galois) iff $F' \not\supseteq K_1$. $\square$

Remark 4.5 (relation with [12], Theorem 10.1). Theorem 10.1 of [12] says: for generically free primitive G -varieties X, Y with X smooth, a G -equivariant correspondence $Y \rightsquigarrow X$ of degree prime to q gives $\text{ed}_{k,q}(X) \geq \text{ed}_{k,q}(Y)$; the proof passes through the q -closure, which is large, and Fact 2.2(b). In the language of Theorem 4.2, a free closed point of degree d on ${}^{\tau_{\text{gen}}}Z$ is a correspondence $W_n \rightsquigarrow Z$ of degree d into the free locus. The prime-to- p part of the profile is thus governed by [12, Thm. 10.1] and collapses to $\text{ed}_{k,p} = 1$ at large d ; the p -power part $\text{ed}^{[p^j]}$ is the analogue for p -power correspondences, for which no large-field mechanism exists, since p -power extensions of K never produce large fields, and no analogue of Theorem 10.1 is available.

5. GLOBAL MINIMA COLLAPSE

Any invariant of τ_{gen} of the form “minimum over all generically free G -varieties Z of dimension $n-1$ of some measure of the free locus of ${}^{\tau_{\text{gen}}}Z$ ” is dominated by its value on the following cheap hypersurface.

Definition 5.1. $S_0 := \{z \in W_n : (\wp z)_{n-1} = 0\} = \wp^{-1}(\{s_{n-1} = 0\}) \subset W_n$.

Proposition 5.2. (a) S_0 is an irreducible G -stable hypersurface of dimension $n-1$ on which G acts freely; ${}^{\tau_{\text{gen}}}W_n \cong \mathbb{A}_K^n$ via $y \mapsto h = y - z$, and under this identification ${}^{\tau_{\text{gen}}}S_0 = \{h \in \mathbb{A}_K^n : (s + \wp h)_{n-1} = 0\}$.

- (b) ${}^{\tau_{\text{gen}}}S_0$ has a closed point of degree p , namely $h = (0, \dots, 0, v)$ with $v^p - v = s_{n-1}$, and no K -point unless $\text{ed}_k(\tau_{\text{gen}}) \leq n-1$. Hence $d_{n-1}(\tau_{\text{gen}}) = \min_Z \delta_{\text{free}}(Z) \leq p$; at $p = 2$ this minimum is 1 if $\text{ed}_k(\mathbb{Z}/2^n) \leq n-1$ and 2 otherwise (the “Ledet indicator”).
- (c) For every $j \geq 1$: $\min\{\text{trdeg}_K F : F/K \text{ finitely generated, } \text{ed}_k(\tau_{\text{gen}} \otimes F) \leq j\} \in \{0, 1\}$; it is 0 iff $\text{ed}_k(\tau_{\text{gen}}) \leq j$.
- (d) Let $\gamma_{\text{free}}(\tau_{\text{gen}}) := \min_Z \min\{\text{geometric genus of a geometrically integral } K\text{-curve } W \subseteq {}^{\tau_{\text{gen}}}Z \text{ whose generic point is free}\}$ over Z as in (b) with $\dim Z = n-1$. Then $\gamma_{\text{free}}(\tau_{\text{gen}}) = 0$ iff $\text{ed}_k(\tau_{\text{gen}}) \leq n-1$, and for $(p, n) = (2, 3)$ one has $\gamma_{\text{free}}(\tau_{\text{gen}}) \leq 1$, witnessed on ${}^{\tau_{\text{gen}}}S_0$ by the elliptic curve $h_2^2 + h_2 = u(u+1)(u+s_1) + s_2$ over K , which has no K -point other than ∞ .

[PROVED (D: COMPUTED)]

Proof. (a) S_0 is G -stable because $\wp$ is G -invariant, and is the Artin–Schreier–Witt cover of $\mathbb{A}^{n-1}$ of class $a := (s_0, \dots, s_{n-2}, 0) \in W_n(K_{(n-1)})$, $K_{(n-1)} = k(s_0, \dots, s_{n-2})$. It is irreducible iff this class has order p^n : $p^{n-1}a = V^{n-1}F^{n-1}a = (0, \dots, 0, s_0^{p^{n-1}})$, and by (5) this is $\equiv 0$ iff $s_0^{p^{n-1}} \equiv s_0 \in \wp K_{(n-1)}$, which fails (s_0 has a simple pole at $s_0 = \infty$ over $k(s_1, \dots, s_{n-2})$). Freeness is inherited from W_n . A K' -point of ${}^{\tau_{\text{gen}}}W_n$ is $y \in W_n(L \otimes_K K')$ with $gy = y + 1$; then $h := y - z$ is G -invariant, i.e. $h \in W_n(K')$, and $y = z + h$; this is the identification with $\mathbb{A}_K^n$ (the K -morphism $\text{Spec } L \times_k W_n \rightarrow \mathbb{A}_K^n$, $(\cdot, y) \mapsto y - z$, is G -invariant and identifies the quotient). The point lies in ${}^{\tau_{\text{gen}}}S_0$ iff $(\wp(z+h))_{n-1} = (s + \wp h)_{n-1} = 0$. (b) By (2) and $\wp V^{n-1} = V^{n-1}\wp$: $s + \wp V^{n-1}(v) = s + V^{n-1}(v^p - v) = (s_0, \dots, s_{n-2}, s_{n-1} + s_{n-1})$, whose last coordinate is 0 for $p = 2$; for odd p take $v^p - v = -s_{n-1}$. The residue field $K(v)$ has degree p ($s_{n-1} \notin \wp K$). Every point of ${}^{\tau_{\text{gen}}}S_0$ is free. A free K -point on the $(n-1)$ -dimensional S_0 gives $\text{ed}_k(\tau_{\text{gen}}) \leq n-1$ (Lemma 2.3); conversely $\text{ed} \leq n-1$ means $d_{n-1} = 1$ by Corollary 4.4 (not necessarily on S_0). The formula $d_{n-1} = \min_Z \delta_{\text{free}}(Z)$ is Corollary 4.4. (c) The value 0 means $\text{ed}_k(\tau_{\text{gen}}) \leq j$. For the bound 1: let $C = \overline{U}_n$ and $F := K({}^{\tau_{\text{gen}}}\overline{U}_n)$, the function field of the twisted curve (geometrically integral, being a twist of the integral curve $\overline{U}_n$). Its generic point is an F -point of ${}^{\tau_{\text{gen}}}\overline{U}_n$ lying in the dense open free locus ${}^{\tau_{\text{gen}}}U_n$, so $\text{ed}_k(\tau_{\text{gen}} \otimes F) \leq 1 \leq j$ by Lemma 2.3, with $\text{trdeg}_K F = 1$. (d) If $\text{ed}_k(\tau_{\text{gen}}) \leq n-1$ there is a compression $\phi: W_n \dashrightarrow Z$ onto a faithful irreducible Z of dimension $n-1$ (Lemma 2.1 and the definition of $\text{ed}_k(W_n)$). Twisting, ${}^{\tau_{\text{gen}}}\phi: \mathbb{A}_K^n \cong {}^{\tau_{\text{gen}}}W_n \dashrightarrow {}^{\tau_{\text{gen}}}Z$

is dominant, and the preimage U of the free locus is a dense open with dense K -points. Take a K -line ℓ through a K -point of U on which $\tau_{\text{gen}}\phi$ is nonconstant (a general line will do); the closure W of its image is a K -unirational curve, hence K -rational (Lüroth over K), with free generic point. Conversely, if $W \subseteq \tau_{\text{gen}}Z$ is a curve with normalisation $\mathbb{P}_K^1$ and free generic point, then $F := K(W) = K(t)$ and $\text{ed}_k(\tau_{\text{gen}} \otimes K(t)) \leq n - 1$ by Lemma 2.3, so $\text{ed}_k(\tau_{\text{gen}}) \leq n - 1$ by Lemma 4.1(i). For $(p, n) = (2, 3)$: on $\tau_{\text{gen}}S_0$ set $h = (0, u, h_2)$; by (3)–(4) (script) the equation $(s + \wp h)_2 = 0$ reads $h_2^2 + h_2 = u^3 + (1 + s_1)u^2 + s_1u + s_2 = u(u + 1)(u + s_1) + s_2$. This is a smooth affine plane curve ($\partial/\partial h_2 = 1$) whose projective closure is an elliptic curve $E'_{(s_1, s_2)}$ (Weierstrass form with $a_1 = 0$, $a_3 = 1$, $\Delta = a_3^4 = 1$), with free generic point since all of $\tau_{\text{gen}}S_0$ is free. Its K -points: over $k(s_1, s_2)$ it is the curve E' of Theorem 6.2 (with (s_0, s_1) renamed (s_1, s_2)), which has only the point ∞ there, and $K = k(s_1, s_2)(s_0)$ is purely transcendental over $k(s_1, s_2)$, so Lemma 4.1(ii) applies. $\square$

Remark 5.3. (b) says that “minimal degree of a free point over all $(n - 1)$ -folds” is exactly $d_{n-1}(\tau_{\text{gen}}) \in \{1, \dots, p\}$: a single number, $\leq p$, and at $p = 2$ a single bit. Likewise (c) shows that the transcendence-degree analogue of the profile carries no information beyond that bit, and (d) that the “minimal genus of a free curve” is 0 or (at $(2, 3)$) 1. Graded information must therefore be sought either *per variety* ($D(Z)$, Section 8) or in the profile of the torsor, which is a base-change invariant and is not dominated by cheap Z : S_0 contributes only $\text{ed}^{[p]} \leq n - 1$, which is Proposition 3.4.

6. CALIBRATION AT $n = 2$

Let $n = 2$, $G_2 = \mathbb{Z}/p^2$, $K_2 = k(s_0, s_1)$, $L_2 = k(z_0, z_1)$, τ_2 the generic torsor.

Proposition 6.1 ([15], Prop. G.1, extended). *Let C be a smooth projective curve with faithful G_2 -action. Then ${}^{\tau_2}C$ has no free K_2 -point; in fact ${}^{\tau_2}C(K_2)$ is in bijection with $C^{G_2}(k)$. Consequently $\text{ed}_k(\mathbb{Z}/p^2) = 2$ and $d_1(\tau_2) \geq 2$.* [PROVED]

Proof. A K_2 -point is an equivariant $f: \text{Spec } L_2 \rightarrow C$; by Lemma 2.4 its image closure Y is a G_2 -stable point (then a fixed k -point, and f is constant) or all of C . In the latter case $k(C) \subseteq L_2 = k(z_0, z_1)$, so $C \cong \mathbb{P}^1$ by Lemma 2.5(i) and $G_2 \hookrightarrow \text{PGL}_2(k)$, contradicting Lemma 2.5(ii). So all K_2 -points are the constant maps to fixed points, none free. Then $d_1(\tau_2) \geq 2$ by Corollary 4.4 (for $d = 1$, $L_2 \otimes K_2 = L_2$ is a field), and $\text{ed}_k(\mathbb{Z}/p^2) = \text{ed}_k(\tau_2) \geq 2$ by Lemma 2.1; ≤ 2 is Proposition 3.4 with $j = 0$. $\square$

Theorem 6.2 ($n = 2$, $p = 2$). *Let $p = 2$. The curve $\bar{U}_2$ is the supersingular elliptic curve $E_0: y_1^2 + y_1 = y_0^3 + y_0^2$ with G_2 acting by $(y_0, y_1) \mapsto (y_0 + 1, y_1 + y_0)$, fixing only ∞ . Its twist is the elliptic curve*

$$E' := {}^{\tau_2}E_0: \quad w^2 + w = u(u + 1)(u + s_0) + s_1 \quad \text{over } K_2 = k(s_0, s_1),$$

and $E'(K_2) = \{\infty'\}$: the Mordell–Weil group of the quartic twist of E_0 by the generic $\mathbb{Z}/4$ -torsor over $k(s_0, s_1)$ is trivial. Every fibre of $u: E' \rightarrow \mathbb{P}_{K_2}^1$ over a point $c \in K_2$ is a free closed point of degree 2, with residue field $K_2(w)$, $w^2 + w = c(c + 1)(c + s_0) + s_1$, and over it $(s_0, s_1) \equiv [s_0 + c^2 + c]$. The profile of τ_2 is

$$\text{ed}^{[1]} = 2, \quad \text{ed}^{[2]} = \text{ed}^{[3]} = 1, \quad \text{ed}^{[d]} = 0 \ (d \geq 4); \quad d_2 = 1, \ d_1 = 2, \ d_0 = 4;$$

$$d_1^{(p')} \in \{3, 5, 7, \dots\} \text{ [OPEN].}$$

[PROVED]

Proof. $U_2 = \{y_0^2 + y_0 = t, \ y_1^2 + y_1 + y_0^3 + y_0^2 = 0\}$ by (4) (the W_2 part), and t is a function of y_0 , so $U_2 \cong E_0^\circ := \{y_1^2 + y_1 = y_0^3 + y_0^2\} \subset \mathbb{A}^2$, a smooth affine curve whose projective closure E_0 is an elliptic curve with one point ∞ at infinity ($a_1 = 0$, $a_3 = 1$, $\Delta = 1$; supersingular since

$a_1 = 0$ in characteristic 2, $j = a_1^{12}/\Delta = 0$). By (3), $y + 1 = (y_0 + 1, y_1 + y_0)$, an automorphism of E_0 without fixed points on E_0° , so ∞ is the unique fixed point and $E_0^{\text{free}} = E_0^\circ = U_2$. By Proposition 5.2(a) (with $n = 2$), ${}^\tau U_2 = \{h = (u, w) \in \mathbb{A}_{K_2}^2 : (s + \wp h)_1 = 0\}$, and by (3)–(4), $(s + \wp h)_1 = s_1 + w^2 + w + u^3 + u^2 + s_0(u^2 + u)$; so ${}^\tau U_2 = E'^\circ := \{w^2 + w = u(u + 1)(u + s_0) + s_1\}$, and ${}^\tau E_0$ is its smooth projective closure E' , with the single point ∞' at infinity (the twist of the fixed point). By Proposition 6.1, $E'(K_2) = {}^\tau E_0(K_2)$ is in bijection with $E_0^{G_2}(k) = \{\infty\}$. The fibre of u over $c \in K_2$ is $\text{Spec } K_2[w]/(w^2 + w - c(c + 1)(c + s_0) - s_1)$; it has no K_2 -point, so it is a field of degree 2, and its point is free (it lies in E'°). Over $K' = K_2(w)$ the vector $h = (c, w)$ satisfies $(s + \wp h)_1 = 0$, so $s + \wp h = [(s + \wp h)_0] = [s_0 + c^2 + c]$.

Profile: $\text{ed}^{[1]} = \text{ed}_k(\mathbb{Z}/4) = 2$; $\text{ed}^{[2]} \leq 1$ by the degree-2 free points (Lemma 2.3), equivalently by Remark 3.5; $\text{ed}^{[d]} \geq 1$ for $d < 4$ and $\text{ed}^{[4]} = 0$ by Proposition 3.2(c). For $d_1^{(p')}$: it is odd and ≥ 3 , finite by Proposition 3.7(b); the exact value is open. $\square$

Proposition 6.3 ($n = 2$, all p). *For every p : $2 \leq d_1(\tau_2) \leq p$, $d_0(\tau_2) = p^2$, and $\text{ed}_k^{[d]}(\tau_2) = 1$ for $p \leq d < p^2$. A degree- p free point of ${}^\tau \bar{U}_2$ lies over $t = s_0$: the fibre of ${}^\tau U_2 \rightarrow \mathbb{A}_{K_2}^1$ over $t = c \in K_2$ is the G_2 -torsor of class $[c] - s$, which for $c = s_0$ equals $V(-s_1)$, so the fibre is $\text{Ind}_{\mathbb{Z}/p}^{\mathbb{Z}/p^2}$ of the Artin-Schreier torsor $v^p - v = -s_1$ and consists of p closed points of degree p .* [PROVED]

Proof. $U_2 \rightarrow \mathbb{A}^1$, $y \mapsto t$, is G_2 -equivariant for the trivial action on $\mathbb{A}^1$, and twists to ${}^\tau U_2 \rightarrow \mathbb{A}_{K_2}^1$. Its fibre over c is the twist of the fibre $\wp^{-1}([c])$, and its K' -points are the $y = z + h$ with $\wp(z + h) = s + \wp h = [c]$, i.e. $h \in W_2(K')$ with $\wp h = [c] - s$: the fibre is the torsor of class $[c] - s$. For $c = s_0$, $[s_0] - s = -V(s_1)$ by (2), the image under V of the $\mathbb{Z}/p$ -class $-s_1 \neq 0$ in $K_2/\wp K_2$; the corresponding torsor is induced from $\mathbb{Z}/p$, i.e. Spec of p copies of $K_2(v)$. These points are free (they lie in ${}^\tau U_2$). So $d_1 \leq p$; $d_1 \geq 2$ is Proposition 6.1, $d_0 = p^2$ is Proposition 3.2(c). $\square$

Question 6.4. *For $p \geq 3$, is $d_1(\tau_2) = p$, i.e. does no extension of $k(s_0, s_1)$ of degree $< p$ (necessarily prime to p) make the generic $\mathbb{Z}/p^2$ -torsor one-dimensional? Equivalently (Corollary 4.4): does every smooth projective faithful G_2 -curve C satisfy $D(C) \cap [2, p-1] = \emptyset$? Note that for $d < p$ prime to p , Proposition 3.7 and the largeness of $K_2^{(p)}$ give no information, since $K_2^{(p)}$ is an infinite union of such extensions. Also open: $d_1^{(p')}(\tau_2)$ for $p = 2$.* [OPEN]

7. THE CASE $(p, n) = (2, 3)$

Let $p = 2$, $n = 3$, $G = \mathbb{Z}/8$, $K = k(s_0, s_1, s_2)$, $L = k(z_0, z_1, z_2)$, $v_i^2 + v_i = s_i$, $z_0^2 + z_0 = s_0$.

Theorem 7.1 (the profile at $(2, 3)$).

$$\text{ed}^{[1]}(\tau_{\text{gen}}) = \text{ed}_k(\mathbb{Z}/8) \in \{2, 3\}, \quad \text{ed}^{[2]} \leq 2, \quad \text{ed}^{[3]} \in \{1, 2\}, \quad \text{ed}^{[4]} = \dots = \text{ed}^{[7]} = 1, \quad \text{ed}^{[8]} = 0;$$

$d_2 \in \{1, 2\}$ (with $d_2 = 1$ iff Ledet's conjecture fails for $\mathbb{Z}/8$), $d_1 \in \{2, 3, 4\}$, $d_0 = 8$, $d_1^{(p')} \geq 3$ finite. Witnesses: $\text{ed}^{[2]} \leq 2$ by $K(v_2)$ (class $(s_0, s_1, 0)$), by $K(v_1)$ (class $(s_0, 0, s_2 + s_1^2 + s_1 v_1)$, Remark 3.5, defined over the transcendence-degree-2 field $k(s_0, s_2 + s_1^2 + s_1 v_1)$) and by $K(z_0)$ (Proposition 3.9); $\text{ed}^{[4]} \leq 1$ by the tower $K(v_1, v')$, $v'^2 + v' = s_2 + s_1^2 + s_1 v_1$ (Lemma 3.3). The degree-2 free points of $\tau_{\text{gen}} X_{0,2}$ (Proposition 8.2) give $\text{ed}^{[2]} \leq 2$ once more. [PROVED]

Proof. $\text{ed}^{[1]}$: [8, Thm. 1.3(b)] (the lower bound 2 is Lemma 2.5(i),(ii) via Corollary 4.4, the upper bound 3 Lemma 2.1). $\text{ed}^{[2]} \leq 2$, $\text{ed}^{[4]} \leq 1$: Proposition 3.4. $\text{ed}^{[d]} \geq 1$ for $d < 8$ and $\text{ed}^{[8]} = 0$: Proposition 3.2(c). The jump degrees follow. The unknown slots are $\text{ed}^{[1]}$, $\text{ed}^{[2]}$ (which is 1 or 2) and $\text{ed}^{[3]}$. $\square$

We now show that the three most natural quadratic extensions, and the naive quartic tower, do not realise $d_1 = 2$ resp. $\text{ed}^{[4]} = 1$.

Lemma 7.2 (curves dominated by a fibration). *Let $\pi: V \rightarrow B$ be a dominant morphism of integral k -varieties whose generic fibre V_η is geometrically integral, and let $\phi: V \dashrightarrow C$ be a dominant rational map to a smooth projective curve. Then either some closed fibre V_b has an irreducible component whose normalisation maps nonconstantly to C , or $k(C) \subseteq k(B)$ (so C is dominated by B). [PROVED]*

Proof. Consider $\phi_\eta: V_\eta \dashrightarrow C_{k(B)}$ over $k(B)$. If ϕ_η is dominant, then after spreading out, $\phi|_{V_b}$ is dominant onto C for all b in a dense open of B , giving the first alternative. Otherwise ϕ_η has image a closed point x of $C_{k(B)}$, and $k(C) \subseteq \kappa(x) \subseteq k(V_\eta) = k(V)$ with $\kappa(x)$ finite over $k(B)$. As V_η is geometrically integral, $k(V)/k(B)$ is a regular extension, so $k(B)$ is algebraically closed in $k(V)$ and $\kappa(x) = k(B)$; hence $k(C) \subseteq k(B)$. $\square$

Theorem 7.3. *Let $(p, n) = (2, 3)$. For each of the fields $K' = K(v_1)$, $K' = K(z_0)$ and $K' = K(v_1, v_2)$ one has*

$$\text{ed}_k(\tau_{\text{gen}} \otimes_K K') = 2.$$

[PROVED]

Proof. Upper bounds: Theorem 7.1 and Proposition 3.9. For $K(z_0)$ the lower bound is Proposition 3.9: $\text{ed}_k(\tau_{\text{gen}} \otimes K_1) \geq \text{ed}_k(\mathbb{Z}/4) = 2$.

$K' = K(v_1)$. First, $L \otimes_K K'$ is a field: the only quadratic subextension of L/K is $K(z_0)$, and $K(v_1) \neq K(z_0)$ because $s_0 + s_1 \notin \wp K$ (pole of order 1 at $s_1 = \infty$); so $K' \cap L = K$. Put $y := v_1 + z_1$; then $y^2 + y = s_1 + z_1^2 + z_1 = z_0^3 + z_0^2$ by (4), so

$$L(v_1) = k(z_0, y, z_1, z_2) = k(E_0 \times \mathbb{A}^2), \quad E_0: y^2 + y = z_0^3 + z_0^2,$$

with z_1, z_2 the coordinates on $\mathbb{A}^2$ (y is algebraic over $k(z_0)$ and $[L(v_1) : L] = 2$ since $z_0^3 + z_0^2 \notin \wp L$, having odd pole order in z_0). Suppose $\text{ed}_k(\tau_{\text{gen}} \otimes K') \leq 1$. By Theorem 4.2 and Corollary 4.4 there is a smooth projective curve C with faithful G -action and a dominant equivariant rational map $\phi: E_0 \times \mathbb{A}^2 \dashrightarrow C$ (the descent field F_0 has $\text{trdeg}_k F_0 = 1$; $\text{trdeg } 0$ is impossible as the torsor over K' is a field, hence non-split). Apply Lemma 7.2 to $\pi = \text{pr}_2: E_0 \times \mathbb{A}^2 \rightarrow \mathbb{A}^2$ (fibres E_0 , geometrically integral): either $E_0 \rightarrow C$ is nonconstant for some fibre, so $g(C) \leq 1$ by Lemma 2.5(iv), and if $g(C) = 1$ then C is isogenous to E_0 (a nonconstant morphism of elliptic curves is an isogeny up to translation), hence supersingular; or $k(C) \subseteq k(\mathbb{A}^2)$, so $C \cong \mathbb{P}^1$ by Lemma 2.5(i). In all cases C is $\mathbb{P}^1$ or a supersingular elliptic curve, and neither carries an automorphism of order 8 (Lemma 2.5(ii),(iii)), contradicting faithfulness. Hence $\text{ed}_k(\tau_{\text{gen}} \otimes K(v_1)) \geq 2$.

$K' = K(v_1, v_2)$. Again $L \otimes_K K'$ is a field ($K' \cap L = K$: K'/K is a $(\mathbb{Z}/2)^2$ -extension whose quadratic subfields are $K(v_1), K(v_2), K(v_1 + v_2)$, of classes $s_1, s_2, s_1 + s_2$, none $\equiv s_0$). With y as before and $y' := v_2 + z_2$, $y'^2 + y' = s_2 + z_2^2 + z_2 = C(z_0, z_1)$ by (4), so $L(v_1, v_2) = k(z_0, y, z_1, y', z_2) = k(\Sigma \times \mathbb{A}_{z_2}^1)$, where

$$\Sigma := \{(z_0, y, z_1, y') : y^2 + y = z_0^3 + z_0^2, \quad y'^2 + y' = C(z_0, z_1)\},$$

$$C(z_0, z_1) = z_1^3 + (z_0^3 + z_0^2 + 1)z_1^2 + (z_0^3 + z_0^2)z_1 + z_0^7 + z_0^4.$$

Σ is an integral surface, fibred over E_0 via (z_0, y) , with fibre over a point of E_0 the affine curve $y'^2 + y' = (\text{monic cubic in } z_1)$, which is smooth and geometrically integral of genus 1 with $j = 0$ (Weierstrass form with $a_1 = 0, a_3 = 1$), hence supersingular. As before, $\text{ed} \leq 1$ would give a dominant equivariant $\Sigma \times \mathbb{A}^1 \dashrightarrow C$. Lemma 7.2 for $\text{pr}_2: \Sigma \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ gives: C is dominated by $\mathbb{A}^1$ (rational), or by Σ . In the latter case apply Lemma 7.2 to $\Sigma \rightarrow E_0$: either a fibre E_{z_0} (supersingular elliptic) maps nonconstantly to C , or E_0 does. Either way C is $\mathbb{P}^1$ or an elliptic curve isogenous to a supersingular one, and Lemma 2.5 excludes a faithful $\mathbb{Z}/8$ -action. $\square$

Remark 7.4 (the argument of [4] via E'' , and isotriviality). [4, §2.6] proves the weaker statement that $\tau_{\text{gen}} \otimes K(v_1)$ is not in Teichmüller form, i.e. $\tau_{\text{gen}} \overline{U}_3$ has no free $K(v_1)$ -point, as follows (with

the class corrected to $(s_0, 0, c)$, $c = s_2 + s_1^2 + s_1 v_1$, Remark 3.5). If $(s_0, 0, c) + \wp(u, w_1, w_2) = (x, 0, 0)$ over $K'' = K(v_1) = k(s_0, v_1, s_2)$, the coordinate of index 1 reads (script)

$$w_1^2 + w_1 = u^3 + u^2 + s_0 u^2 + s_0 u = u(u+1)(u+s_0),$$

so $(u, w_1) \in E''(K'')$ for the elliptic curve $E'': w^2 + w = u(u+1)(u+s_0)$ over $k(s_0)$. Here E'' has $a_1 = 0$, hence $j(E'') = 0$: it is *isotrivial*, a twist of the constant curve $y^2 + y = x^3$. The note asserts $E''(K'') = E''(k(s_0))$ because K'' is purely transcendental over $k(s_0)$; this is correct, but not for any reason involving non-isotriviality or Lang–Néron: it is Lemma 4.1(ii), valid for every elliptic curve. So $u, w_1 \in k(s_0)$, and the coordinate of index 2 becomes $w_2^2 + w_2 = c + \Phi(s_0, u, w_1)$ with Φ the polynomial displayed by the script, i.e. $w_2^2 + w_2 = s_2 + \psi$ with $\psi \in k(s_0, v_1)$. This has no solution in $K'' = k(s_0, v_1)(s_2)$: the right-hand side has a simple pole at $s_2 = \infty$, while $w_2^2 + w_2$ has even or non-negative valuation there. Hence no free $K(v_1)$ -point on ${}^{\tau_{\text{gen}}}\bar{U}_3$. Theorem 7.3 strengthens this to “no free $K(v_1)$ -point on the twist of *any* faithful G -curve”.

Proposition 7.5 (shape of a hypothetical quadratic witness on $\bar{U}_3$). *Let E' be the elliptic curve of Theorem 6.2 over $K_2 = k(s_0, s_1)$, base-changed to $K = K_2(s_2)$; $E'(K) = \{\infty'\}$ by Lemma 4.1(ii). The truncation $\bar{U}_3 \rightarrow \bar{U}_2 = E_0$ twists to a degree-2 morphism $\rho: {}^{\tau_{\text{gen}}}\bar{U}_3 \rightarrow E'$, and ${}^{\tau_{\text{gen}}}\bar{U}_3(K) = \{\infty''\}$. Let $K' = K(v)$, $v^2 + v = f \in K \setminus \wp K$, be a quadratic extension. Then ${}^{\tau_{\text{gen}}}\bar{U}_3$ has a free K' -point (equivalently: $s \equiv [t]$ over K' for some $t \in K'$) if and only if there is $Q = (u, w_1) \in E'(K')$ with*

- (i) $u \notin K$ (in particular $Q \notin E'(K)$ and Q does not lie in a u -fibre over a K -point), and
- (ii) $e_Q := s_2 + \Phi_2(s_0, s_1, u, w_1) \in \wp K'$, where Φ_2 is the polynomial with $(s + \wp(u, w_1, w_2))_2 = w_2^2 + w_2 + e_Q$.

Condition (i) says that the K_2 -morphism from the curve $D_f: v^2 + v = f$ (a curve over K_2 with function field K') to E' defined by Q does not have u pulled back from $D_f \rightarrow \mathbb{P}_{s_2}^1$. [PROVED]

Proof. A free K' -point is $h = (u, w_1, w_2) \in W_3(K')$ with $(s + \wp h)_1 = (s + \wp h)_2 = 0$ (Proposition 5.2(a) and $U_3 = \{(\wp y)_1 = (\wp y)_2 = 0\}$); the first equation says $(u, w_1) \in E'(K')$ (Theorem 6.2), the second is (ii) with e_Q as defined ((3)–(4) show $(s + \wp h)_2 = w_2^2 + w_2 + s_2 + \Phi_2$ with Φ_2 independent of w_2). ${}^{\tau_{\text{gen}}}\bar{U}_3(K) = \{\infty''\}$ by the argument of Proposition 6.1 with L in place of L_2 ($\bar{U}_3$ has the unique fixed point ∞). It remains to show that a free K' -point forces $u \notin K$. Suppose $u \in K$. Then $t := s_0 + u^2 + u \in K$, and over K' we have $s \equiv [t]$ with $t \in K$. Consider the inflation–restriction sequence for the quadratic extension K'/K with trivial action on $\mathbb{Z}/8$:

$$0 \rightarrow H^1(\text{Gal}(K'/K), \mathbb{Z}/8) \rightarrow H^1(K, \mathbb{Z}/8) \rightarrow H^1(K', \mathbb{Z}/8), \quad H^1(\text{Gal}(K'/K), \mathbb{Z}/8) \cong \mathbb{Z}/2,$$

generated by the class of the $\mathbb{Z}/8$ -torsor induced from K'/K along $\mathbb{Z}/2 \hookrightarrow \mathbb{Z}/8$, whose Witt class is $V^2(f) = (0, 0, f)$ (Section 2.2). Since s and $[t]$ both come from $W_3(K)$ and agree over K' , $s - [t] \equiv \varepsilon(0, 0, f)$ over K with $\varepsilon \in \{0, 1\}$. If $\varepsilon = 0$, $s \equiv [t]$ over K , a free K -point, excluded. If $\varepsilon = 1$, truncating to W_2 gives $(s_0, s_1) \equiv (t, 0)$ over K , i.e. a K -point (u, w) of E' with u as given, whence a free K -point of E' , excluded by $E'(K) = \{\infty'\}$. $\square$

Remark 7.6. For $p = 2$, $n = 2$ the same inflation–restriction argument shows that a quadratic $K' = K_2(v)$ over which τ_2 becomes Teichmüller with $t \in K_2$ must have $v^2 + v \equiv$ the correction class; this is consistent with Theorem 6.2, where $K' = K_2(w)$, $w^2 + w = c(c+1)(c+s_0) + s_1$, and $(s_0, s_1) - [s_0 + c^2 + c] \equiv (0, c(c+1)(c+s_0) + s_1)$. At $n = 3$ the analogous “ $u \in K$ ” witnesses are exactly what Proposition 7.5 rules out: the two upper layers cannot both be killed by one quadratic extension of this shape. A witness with $u \notin K$ requires a point of E' over the function field of D_f (a curve over K_2) not coming from the s_2 -line, i.e. a non-constant K_2 -morphism $D_f \rightarrow E'$ not factoring through $\mathbb{P}_{s_2}^1$ in the u -coordinate, and in addition the Artin–Schreier condition (ii). Since $E'(K_2(s_2)) = \{\infty'\}$, such a point is a point of positive order on the quadratic twist of E' by D_f , viewed over $K_2(s_2)$, or a point defined over the quadratic extension in another way; Algorithm 9.1 makes this searchable.

8. PER-SURFACE FREE DEGREES AND $X_{0,2}$

Let X be a smooth projective surface with a faithful G -action, $G = \mathbb{Z}/p^3$, and $D(X) = D(X, \tau_{\text{gen}})$ as in Definition 4.3.

- Proposition 8.1.** (a) $1 \in D(X)$ iff X is τ_{gen} -versal iff there is a G -compression $W_3 \dashrightarrow X$ [15, Prop. B.2]. So $\delta_{\text{free}}(X) \geq 2$ for every X iff $\text{ed}_k(\mathbb{Z}/p^3) = 3$ [15, Thm. D.1].
 (b) If X has a G -fixed k -point (automatic for compression targets [15, Prop. C.1]), then $\tau_{\text{gen}}X(K) \neq \emptyset$, $D(X) \neq \emptyset$ and $\gcd D(X) = 1$. (Whether $D(X)$ contains all large integers, as asserted in [4, Prop. 3.2(b)], is not proved here.)
 (c) $d_2(\tau_{\text{gen}}) \leq p$, and $d_2(\tau_{\text{gen}}) = \min_Z \delta_{\text{free}}(Z)$ over quasi-projective generically free primitive G -surfaces Z (Corollary 4.4, Proposition 5.2); the minimum over smooth projective faithful G -surfaces equals 1 iff $\text{ed}_k(\mathbb{Z}/p^3) \leq 2$, by (a).

[(A) IMPORTED, (B),(C) PROVED]

Proof. (a) is [15]. (b) The fixed point gives the constant map, a smooth K -point P' of $\tau_{\text{gen}}X$ (Fact 2.2(a)); $\tau_{\text{gen}}X$ is a smooth projective geometrically integral K -surface. A general hyperplane section W through P' is a geometrically integral K -curve (Bertini over the infinite field K) not contained in the closed non-free locus; let $\tilde{W}$ be its normalisation, of genus γ , with the K -point $\tilde{P}'$ over P' . For $d \geq 2\gamma + 1$ the linear system $|d\tilde{P}'|$ is base-point free of dimension ≥ 1 , so it contains effective K -rational divisors of degree d avoiding any given finite set, in particular supported in the free locus; the degrees of the closed points in the support of such a divisor sum to d . Doing this for d and $d + 1$ shows $D(X) \neq \emptyset$ and that $\gcd D(X)$ divides $\gcd(d, d + 1) = 1$. (c) is a restatement of Corollary 4.4 and Proposition 5.2(b), and of (a) for the value 1. $\square$

Proposition 8.2 ($X_{0,2}$). Let $p = 2$ and $X = X_{0,2} \subset \mathbb{P}^2 \times \mathbb{P}^2$, $\sum x_i^2 y_i = \sum x_i y_i^2 = 0$, with $g(x, y) = (My, Nx)$ as in [8, §1.4], an automorphism of order 8 fixing $P_0 = ((0:0:1), (1:0:0))$. Then $1 \notin D(X)$ and $2 \in D(X)$: $\delta_{\text{free}}(X_{0,2}) = 2$.
 [IMPORTED + PROVED]

Proof. $1 \notin D(X)$ is the theorem that $(X_{0,2}, g)$ is not versal [8, Thm. 1.9(iv)], [7, Thms. 17.3, 17.5] together with (a) above. For $2 \in D(X)$: let $b = x_1/x_0 + y_1/y_2$. With $M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$, $N = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ one computes $(My)_0 = y_2$, $(My)_1 = y_1$, $(Nx)_1 = x_0 + x_1$, $(Nx)_2 = x_0$, so $g^*b = y_1/y_2 + (x_0 + x_1)/x_0 = b + 1$. Thus $b: X \dashrightarrow \mathbb{A}^1 = W_1$ is equivariant for the translation action of G through $G/\langle g^2 \rangle \cong \mathbb{Z}/2$, and twists to a K -rational map $\tau_{\text{gen}}X \dashrightarrow \tau_{\text{gen}}\mathbb{A}^1 \cong \mathbb{A}_K^1$ (Proposition 5.2(a) with $n = 1$ and the class s_0). The fibre over a K -point β of the twisted line is, geometrically, the member $\{b = \beta'\}$ of the pencil with $\beta' \in L$, $\beta' = z_0 + \kappa$, $\kappa \in K$, transcendental over k ; by [7, Rem. 17.4] (imported) the members of the pencil $\{b = \beta'\}$, β' generic, are geometrically integral of geometric genus 2. Let $\tilde{W}$ be the normalisation of this fibre, a smooth geometrically integral curve of genus 2 over K . Its canonical system is a K -rational base-point-free g_2^1 , $\tilde{W} \rightarrow \mathbb{P}_K^1$. The fibres over all but finitely many K -points of $\mathbb{P}_K^1$ are effective K -divisors of degree 2 supported in the free locus (the non-free locus meets W in finitely many points). Such a divisor is either two K -points or one closed point of degree 2; the former would be free K -points of $\tau_{\text{gen}}X$, excluded. Hence $2 \in D(X_{0,2})$. $\square$

Remark 8.3 (a g -invariant elliptic fibration on $X_{0,2}$). The scripts `inter_ed/nsg_isotropy.py`, `nsg_fibration.py` of [4] compute, on the 42 lines/points curves generating $\text{NS}(X_{0,2})$, the seven g -orbits (sizes 2, 4, 4, 8, 8, 8, 8) and their Gram matrix, and find two size-8 orbits of self-intersection 0 forming I_8 -cycles, giving a g -invariant elliptic fibration $X_{0,2} \rightarrow \mathbb{P}^1$ with fibres I_8, I_8, I_1^* , on which g acts fibrewise with order 8 on every smooth fibre. Twisting, $\tau_{\text{gen}}X_{0,2} \rightarrow \mathbb{P}_K^1$ has geometrically integral genus-1 fibres over K -points, each with free generic point, and none of these fibres has a K -point (since $\tau_{\text{gen}}X_{0,2}(K)$ is the single point P' on the I_1^* fibre). This is the $n = 3$ analogue of $E'(K_2) = \{\infty'\}$, and it makes $\gamma_{\text{free}}(X_{0,2}) = 1$ in the per-surface sense of [4, §4]. We have not re-verified the lattice computation here.
 [COMPUTED IN [4]]

9. WHY THE PROFILE IS INTERMEDIATE; OPEN QUESTIONS; THE FIRST COMPUTATION

9.1. Intermediate. Over every large field $F \supseteq K$ one has $d_1(\tau_{\text{gen}} \otimes F) = 1$ (Fact 2.2(d)): the profile of $\tau_{\text{gen}} \otimes F$ is $(\leq 1, \dots)$ and carries nothing. Over K itself, $\text{ed}^{[1]}(\tau_{\text{gen}})$ is the Ledet indicator, but the slots $\text{ed}^{[d]}$ for $1 < d < p^{n-1}$ are neither the indicator nor determined by any large-field or cohomological invariant ($\text{cd}_p(K) \leq 1$; index and elementary obstruction of $\tau_{\text{gen}}X$ are trivial as $\tau_{\text{gen}}X(K) \neq \emptyset$ for every candidate). At $(2, 3)$ the pair $(\text{ed}^{[1]}, d_1) \in \{2, 3\} \times \{2, 3, 4\}$ has, a priori, several values, and the statements “ $d_1 = 4$ ” (no single quadratic or cubic extension makes the generic $\mathbb{Z}/8$ -torsor one-dimensional; the two upper Artin–Schreier layers must be killed separately) and “ $d_1 = 2$ ” are both consistent with everything known, including the theorem $\text{ed}_{k,p}(\mathbb{Z}/8) = 1$. Theorem 7.3 shows that the obvious quadratic and quartic witnesses fail, in a way that is again invisible over large fields.

9.2. Open questions.

- (Q1) $d_1(\tau_{\text{gen}})$ at $(p, n) = (2, 3)$: is $\text{ed}^{[2]}(\tau_{\text{gen}}) = 1$? By Corollary 4.4 and Theorem 4.2: is there a smooth projective faithful $\mathbb{Z}/8$ -curve C and a quadratic K'/K , $K' \neq K(z_0)$, with a free K' -point on $\tau_{\text{gen}}C$? On $C = \overline{U}_3$ the answer is governed by Proposition 7.5.
- (Q2) $\text{ed}^{[3]}(\tau_{\text{gen}})$ at $(2, 3)$: does a cubic extension of K suffice? Cubic extensions are prime to p , so this is the first slot of the prime-to- p tail: $d_1^{(p')}(\tau_{\text{gen}}) \in \{3, 5, 7, \dots\}$.
- (Q3) $n = 2$, $p \geq 3$: is $d_1(\tau_2) = p$ (Question 6.4)? And $d_1^{(p')}(\tau_2)$ for all p .
- (Q4) General n : is $\text{ed}^{[p^j]}(\tau_{\text{gen}}) = n - j$, i.e. are the truncation towers optimal among p -power extensions? Even $\text{ed}^{[p]}(\tau_{\text{gen}}) \geq n - 1$ for $n = 3$, $p = 2$ is open (it is Q1 for $j = 1$ together with Ledet).

9.3. The first computation.

Algorithm 9.1 (deciding whether $\overline{U}_3$ admits a quadratic witness at $(2, 3)$). Input: a class of candidate $f \in K = k(s_0, s_1, s_2)$, e.g. $f \in k(s_0, s_1)[s_2]$ of bounded degree. Output: whether $\tau_{\text{gen}}\overline{U}_3$ has a free $K(v)$ -point, $v^2 + v = f$.

- (1) (Witt data.) Use (3)–(4) to write the two equations of a free point $h = (u, w_1, w_2) \in W_3(K')$: $w_1^2 + w_1 = u(u+1)(u+s_0)+s_1$ and $w_2^2 + w_2 = e_Q = s_2 + \Phi_2(s_0, s_1, u, w_1)$ (script `witt_p2n3.py` prints Φ_2 as the coordinate-2 polynomial).
- (2) (Points of E' over K' .) $K' = K_2(s_2)(v)$ is the function field of the curve D_f over $K_2 = k(s_0, s_1)$. Points $Q \in E'(K')$ are K_2 -morphisms $D_f \rightarrow E'$. Parametrise: for f with D_f of genus 0 over K_2 (e.g. $f = s_2$, $f = s_2^2 + s_2 \cdot a$, or any f with a K_2 -rational point on D_f and $\deg_{s_2}$ reduced to ≤ 1), K' is a rational function field $K_2(\theta)$ and $E'(K') = E'(K_2) = \{\infty'\}$ by Lemma 4.1(ii): *no witness*. So only f with D_f of positive genus can work; for f with $\deg_{s_2} f = 3$ (after Artin–Schreier reduction) D_f is a genus-1 curve over K_2 and $E'(K')$ contains $\text{Hom}_{K_2}(D_f, E')$; if D_f has a K_2 -point it is an elliptic curve and $\text{Hom}(D_f, E')$ is the group of isogenies (up to translation), computable from j -invariants and the supersingularity of both; for $\deg_{s_2} f \geq 5$ one needs $\text{Hom}_{K_2}(D_f, E')$ for hyperelliptic D_f , i.e. elliptic subcovers, which is a finite computation for each f via the decomposition of $\text{Jac}(D_f)$ or, more concretely, by searching for $u, w_1 \in K_2(s_2)[v]$ of bounded height satisfying the first equation with $u \notin K$.
- (3) (Condition (i).) Discard Q with $u \in K$ (Proposition 7.5).
- (4) (Artin–Schreier membership.) For each remaining Q test $e_Q \in \wp K'$: write $K' = K_2(D_f)$, reduce e_Q at every place of D_f/K_2 by subtracting $w^2 + w$ to make all pole orders odd (the standard Artin–Schreier reduction), and test whether the reduced element is 0; a pole of odd order at any place certifies $e_Q \notin \wp K'$.

- (5) (Conclusion.) A surviving Q gives $d_1(\tau_{\text{gen}}) = 2$ with explicit $t = s_0 + u^2 + u \in K'$, $s \equiv [t]$ over K' . If no f of the class survives, the class is excluded; excluding *all* f (i.e. proving $d_1 \neq 2$ on $\overline{U}_3$) needs a structural argument, e.g. that $E' \otimes K'$ has no point with $u \notin K$ for any quadratic K' , which is a statement about $\text{Hom}_{K_2}(D_f, E')$ for all Artin–Schreier curves D_f over K_2 and is where Mordell–Weil input (ranks of quadratic twists of E' over $K_2(s_2)$) enters.

The same script, applied to the genus-2 pencil of Proposition 8.2, decides $\delta'_{\text{free}}(X_{0,2}) := \min(D(X_{0,2}) \setminus 2\mathbb{Z})$, expected in $\{3, 5\}$ [4, §3.4].

Remark 9.2 (what has been proved, what is imported). Proved here from the quoted statements of [12], [3] (via [12]), Artin–Schreier–Witt theory and standard curve theory: Propositions 3.2, 3.4, 3.6, 3.7, 3.9, Lemmas 3.3, 4.1, 7.2, Theorems 4.2, 6.2, 7.1, 7.3, Propositions 5.2, 6.1, 6.3, 7.5, and the degree-2 point of Proposition 8.2 modulo the imported genus-2 statement. Imported without re-verification: Lemma 2.1 and the statements of [15] quoted in Proposition 8.1(a), the non-versality of $(X_{0,2}, g)$ and the genus of the b -pencil [7], and the lattice computation of Remark 8.3. Flagged: Proposition 8.1(b) proves $\gcd D(X) = 1$ but not that all large integers lie in $D(X)$; Proposition 3.7(a) is stated with the caveat of Remark 3.8.

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10. THE PROFILE IN ARBITRARY CHARACTERISTIC, AND THE CHARACTERISTIC-ZERO PICTURE

Addendum (27 September 2026). Sections 1–9 treat $G = \mathbb{Z}/p^n$ in characteristic p . This section records what the definition of the profile gives for an arbitrary finite group G over an arbitrary field k , and what it looks like in characteristic 0 (more generally, in characteristic prime to $|G|$), where the mechanisms are entirely different: instead of Artin–Schreier layers, the profile is governed by Sylow divisibility (Theorem 10.10) and by the roots-of-unity defect $\mathrm{ed}_k(G) - \mathrm{ed}_{k(\mu_e)}(G)$ (Corollary 10.5). Statements carry the tags of Section 1; in addition, §10.6 says explicitly which statements are rewrites of results in the literature and which are new. As in the rest of the note, literature is quoted without re-verification and the corresponding statements are tagged [cited].

10.1. Conventions. k is any field, G a finite group (constant group scheme over k), $F \supseteq k$ a field, $\tau: T \rightarrow \mathrm{Spec} F$ a G -torsor. The classifying homomorphism $\varphi_\tau: \Gamma_F \rightarrow G$ is defined up to conjugation, its image $H = \mathrm{im} \varphi_\tau$ up to conjugacy, and $T \cong \coprod_{G/H} \mathrm{Spec} E$ with E/F Galois with group H (the *splitting field* of τ). We use, in every characteristic: (B1) $\mathrm{ed}_k(\tau \otimes_F F') \leq \mathrm{ed}_k(\tau)$; (B2) $\mathrm{ed}_k(\tau) \geq \mathrm{ed}_{k'}(\tau)$ for $k \subseteq k' \subseteq F$; (B3) $\mathrm{ed}_k(\tau) = 0$ iff τ descends to the algebraic closure of k in F (for k algebraically closed: iff τ is split; for general k this is *not* “descends to k ”: $k = \mathbb{Q}$, $F = \mathbb{Q}(i)(t)$, $\tau = F(\sqrt{1+i})$ has $\mathrm{ed}_{\mathbb{Q}} = 0$ but is not $F(\sqrt{a})$ with $a \in \mathbb{Q}$, since $N_{\mathbb{Q}(i)/\mathbb{Q}}(1+i) = 2$ is not a square); (B4) Lemma 4.1(i), purely transcendental base change does not change ed_k . For a prime q , $\mathrm{ed}_k(\tau; q) = \min\{\mathrm{ed}_k(\tau \otimes F') : F'/F \text{ finite separable, } q \nmid [F' : F]\}$ and $\mathrm{ed}_k(G; q) = \sup_\tau \mathrm{ed}_k(\tau; q)$ [31, (2.5)]. The generic torsor $\tau_{\mathrm{gen}}: \mathrm{Spec} L \rightarrow \mathrm{Spec} K$, $L = k(V)$, $K = L^G$, is taken for a faithful representation V (or any faithful generically free G -variety with $k(V)/k$ regular); $\mathrm{ed}_k(\tau_{\mathrm{gen}}) = \mathrm{ed}_k(G)$ [18, 29]. The profile $\mathrm{ed}_k^{[d]}(\tau)$ and the jump degrees $d_j(\tau)$ are as in Definition 3.1.

10.2. Formal properties, valid in every characteristic.

Proposition 10.1 (shape). *Let τ be a G -torsor over $F \supseteq k$.*

- (a) $d \mapsto \mathrm{ed}_k^{[d]}(\tau)$ is non-increasing with values in $\{0, \dots, \mathrm{ed}_k(\tau)\}$, $\mathrm{ed}_k^{[1]}(\tau) = \mathrm{ed}_k(\tau)$, and $d_{\mathrm{ed}(\tau)} = 1 \leq d_{\mathrm{ed}(\tau)-1} \leq \dots \leq d_0 \leq |\mathrm{im} \varphi_\tau|$.
- (b) If $[F' : F] = e$ and $d \geq 1$ then $\mathrm{ed}_k^{[de]}(\tau) \leq \mathrm{ed}_k^{[d]}(\tau \otimes F') \leq \mathrm{ed}_k^{[d]}(\tau)$; hence $d_j(\tau) \leq [F' : F] d_j(\tau \otimes F')$ for every finite F'/F and every j .
- (c) $\mathrm{ed}_k^{[d]}(\tau) = \min\{j : d_j(\tau) \leq d\}$: the jump degrees determine the profile.
- (d) If $k \subseteq k' \subseteq F$ then $\mathrm{ed}_{k'}^{[d]}(\tau) \leq \mathrm{ed}_k^{[d]}(\tau)$ for all d .
- (e) $\mathrm{ed}_k^{[d]}(\tau) = 0$ iff some F'/F of degree $\leq d$ makes $\tau \otimes F'$ descend to the algebraic closure of k in F' ; for k algebraically closed, iff τ is split by an extension of degree $\leq d$.

[proved]

Proof. (a) Over the splitting field E , $[E : F] = |H|$, the torsor is split, so $d_0 \leq |H|$; the rest is (B1). (b) is Proposition 3.2(b) verbatim; if $[F'' : F] = d_j(\tau \otimes F')$ realises $\mathrm{ed}(\tau \otimes F'') \leq j$ then $[F'' : F] = [F' : F] d_j(\tau \otimes F')$. (c) is a restatement, (d) is (B2) applied to each $\tau \otimes F'$, (e) is (B3). $\square$

Proposition 10.2 (the last jump degree). *Let $H = \mathrm{im} \varphi_\tau$ and E the splitting field of τ .*

- (a) $d_0(\tau) \leq \min\{[F' : F] : \tau \otimes F' \text{ split}\} = |H|$, with equality if k is algebraically closed.
- (b) Let k be perfect and E/k regular. If $[F' : F] < \infty$ and $\tau \otimes F'$ descends to the algebraic closure k_1 of k in F' , then $[F' : F] \geq |H| \cdot [k_1 : k]$. In particular $d_0(\tau) = |H|$.

- (c) For τ_{gen} over a perfect k : $d_0(\tau_{\text{gen}}) = |G|$, and $\text{ed}_k(\tau_{\text{gen}} \otimes F') = 0$ forces $[F' : K] \geq |G| [k_1 : k]$.

[proved]

Proof. (a) $T \otimes_F F'$ is split iff E embeds F -linearly into F' , which forces $[F' : F] \geq |H|$; $F' = E$ works; for k algebraically closed use (B3). (b) Let $\tau \otimes F' \cong \tau_0 \otimes_{k_1} F'$ with τ_0 over k_1 , of splitting field l/k_1 Galois with group H_0 . Comparing total spaces inside an algebraic closure of F' we may arrange $EF' = lF'$. Since l/k is finite separable and E/k regular, $l \otimes_k E$ is a field and $[lE : E] = [l : k] = |H_0| [k_1 : k]$; as $lE \subseteq EF'$,

$$|H_0| [k_1 : k] |H| = [lE : E] [E : F] \leq [EF' : F] = [lF' : F'] [F' : F] \leq |H_0| [F' : F].$$

- (c) $E = L$ is regular over k and $H = G$. □

Remark 10.3. Without regularity $d_0 < |H|$ is possible ($k = \mathbb{Q}$, $\tau = \mathbb{Q}(t)(\sqrt{2})$ is constant, $d_0 = 1$). Proposition 10.2(c) is the general form of Proposition 3.2(c). For k algebraically closed, $\text{ed}^{[d]}(\tau) \geq 1$ for $d < |H|$: the last slot of the profile is the only one understood in general.

Proposition 10.4 (base-field extension bound). *Let k'/k be finite of degree d . Then $\text{ed}_k^{[d]}(\tau) \leq \text{ed}_{k'}(G)$. More precisely, every field factor F' of $F \otimes_k k'$ has $[F' : F] \leq d$ and $\text{ed}_k(\tau \otimes F') \leq \text{ed}_{k'}(\tau \otimes F') \leq \text{ed}_{k'}(G)$.* [proved]

Proof. $F' \supseteq k'$ and $[F' : F] \leq d$. A descent $\tau \otimes F' \cong \tau_0 \otimes_{F_0} F'$ with $k' \subseteq F_0 \subseteq F'$ and $\text{trdeg}_{k'} F_0 \leq \text{ed}_{k'}(G)$ is also a descent over k , and $\text{trdeg}_k F_0 = \text{trdeg}_{k'} F_0$ since k'/k is algebraic. □

Corollary 10.5 (roots-of-unity defect). *Let e be the exponent of G , $\text{char } k \nmid e$, and $d_e = [k(\mu_e) : k]$. For every G -torsor τ over every $F \supseteq k$,*

$$\text{ed}_k^{[[F(\mu_e):F]]}(\tau) \leq \text{ed}_k^{[d_e]}(\tau) \leq \text{ed}_{k(\mu_e)}(G).$$

In particular: (i) if G is abelian, $\text{ed}_k^{[d_e]}(\tau) \leq \text{rank } G$ [19]; (ii) if G is a p -group and $\text{char } k \neq p$, $\text{ed}_k^{[p-1]}(\tau) \leq \text{ed}_k^{[[k(\mu_p):k]]}(\tau) \leq \text{ed}_{k(\mu_p)}(G)$, the minimal dimension of a faithful representation of G over $k(\mu_p)$ [26]. [proved; the values are cited]

Remark 10.6. For τ_{gen} and $F' = K(\mu_e)$ the torsor $\tau_{\text{gen}} \otimes F'$ is the generic torsor over $k(\mu_e)$, so $\text{ed}_{k(\mu_e)}(\tau_{\text{gen}} \otimes K(\mu_e)) = \text{ed}_{k(\mu_e)}(G)$ exactly, but whether ed_k of this torsor equals $\text{ed}_{k(\mu_e)}(G)$ is not known to us in general. [open]

10.3. Sylow divisibility of the jump degrees.

Lemma 10.7 (q -closure). *Let q be a prime, $S \leq \Gamma_F$ a Sylow pro- q subgroup and $F^{(q)} = (F^{\text{sep}})^S$. Then $F^{(q)}$ is a union of finite subextensions of degree prime to q , every finite separable extension of $F^{(q)}$ has q -power degree, and $\text{ed}_k(\tau; q) = \text{ed}_k(\tau \otimes F^{(q)})$ [31, (2.5)]. If F'/F is finite separable with $q \nmid [F' : F]$, then there is an F -embedding $F' \hookrightarrow F^{(q)}$, and hence $\text{ed}_k(\tau \otimes F') \geq \text{ed}_k(\tau; q)$.* [cited; last statement proved]

Proof. $F' \otimes_F F^{(q)}$ is a product of finite separable extensions of $F^{(q)}$, each of q -power degree, whose degrees sum to $[F' : F]$; if none had degree 1 the sum would be divisible by q . So some factor is $F^{(q)}$, giving the embedding, and (B1) gives the inequality. (For a *given* embedding $F' \subseteq F^{\text{sep}}$ the compositum $F'F^{(q)}$ need not be $F^{(q)}$: $F = \mathbb{Q}$, $q = 2$, $F' = \mathbb{Q}(\sqrt[3]{2})$ has $F' \otimes_{\mathbb{Q}} \mathbb{Q}^{(2)} \cong \mathbb{Q}^{(2)} \times \mathbb{Q}^{(2)}(\sqrt{-3})$.) □

Lemma 10.8 (purely inseparable invariance). *Let $\text{char } k = p$, k perfect, G any finite group, F'/F finite purely inseparable. Then $\text{ed}_k(\tau \otimes F') = \text{ed}_k(\tau)$; consequently $\text{ed}_k^{[d]}(\tau)$ is the minimum over separable F'/F of degree $\leq d$.* [proved]

Proof. As in Proposition 3.6, with $\sigma: F^{1/p^m} \rightarrow F$, $x \mapsto x^{p^m}$; the p^m -power map extends to $F^{1/p^m} F^{\text{sep}} \rightarrow F^{\text{sep}}$ and commutes with all automorphisms, so the induced isomorphism $\Gamma_{F^{1/p^m}} \rightarrow \Gamma_F$ is the canonical one and the classifying homomorphism of $\sigma_*(\tau \otimes F^{1/p^m})$ is φ_τ . This recovers Proposition 3.6 without Witt vectors. $\square$

Proposition 10.9. *For every prime q , $\text{ed}_k(\tau_{\text{gen}}; q) = \text{ed}_k(G; q)$. [cited: [31, 29]; sketch below]*

Sketch. “ $\leq$ ” is the definition. For “ $\geq$ ” let $e = \text{ed}_k(\tau_{\text{gen}}; q)$, realised by a prime-to- q extension K'/K and a descent to K_0 , $\text{trdeg}_k K_0 \leq e$; spread out to a G -torsor $Z \rightarrow B$, $\dim B \leq e$, and to a G -equivariant rational map $W' \dashrightarrow Z$ from the normalisation W' of V in $L \otimes_K K'$, finite of degree $[K' : K]$ over V and étale over a dense G -stable open $V_1 \subseteq V$. Given any torsor τ over an infinite F , the twist ${}^\tau V$ is an affine space over F , so a free F -point in ${}^\tau V_1$ exists and gives an equivariant $f: T \rightarrow V_1$; the pullback $T \times_V W' \rightarrow T$ is finite étale of degree $[K' : K]$, hence $\cong T \times_F A$ for a finite étale F -algebra A of that degree. Some field factor F' of A has $q \nmid [F' : F]$ and $T_{F'} \rightarrow W' \dashrightarrow Z$ is equivariant into a free G -variety, so $\text{ed}_k(\tau \otimes F') \leq \dim B \leq e$ by Lemma 2.3 (valid in any characteristic, Theorem 10.13). $\square$

Theorem 10.10 (Sylog divisibility). *Let $\text{char } k = 0$, or k perfect (then use Lemma 10.8). Let τ be a G -torsor over F .*

- (a) *For every prime q and every $j < \text{ed}_k(\tau; q)$, q divides $d_j(\tau)$: every F'/F with $\text{ed}_k(\tau \otimes F') \leq j$ has $q \mid [F' : F]$.*
- (b) *Every prime q with $\text{ed}_k(G; q) > j$ divides $d_j(\tau_{\text{gen}})$; hence $d_j(\tau_{\text{gen}}) \geq \prod_{q: \text{ed}_k(G; q) > j} q$.*
- (c) *For every $d \geq 1$, $\text{ed}_k^{[d]}(\tau_{\text{gen}}) \geq \min_{1 \leq d' \leq d} \max_{q \nmid d'} \text{ed}_k(G; q) \geq \max_{q > d} \text{ed}_k(G; q)$ (q prime).*
- (d) *The prime-to- q profile $\text{ed}_k^{[d]'}(\tau) = \min\{\text{ed}_k(\tau \otimes F') : [F' : F] \leq d, q \nmid [F' : F]\}$ is non-increasing with terminal value $\text{ed}_k(\tau; q)$, reached at a finite prime-to- q degree; for τ_{gen} the terminal value is $\text{ed}_k(G; q)$.*

[proved, granting Proposition 10.9]

Proof. (a) is Lemma 10.7; (b) adds Proposition 10.9; (c) a witness for $\text{ed}^{[d]}$ has some degree $d' \leq d$ and (a) applies to every $q \nmid d'$; (d) is Lemma 10.7 together with the finiteness argument of Proposition 3.7(a). $\square$

Remark 10.11. (i) In the language of Theorem 4.2 and Remark 4.5, (a) is Reichstein–Scavia’s Theorem 10.1 applied to the correspondence of degree prime to q furnished by a free closed point; the proof above via the q -closure is the same mechanism. (ii) Consistency with Sections 3–9: for $G = \mathbb{Z}/p^n$ in characteristic p , $\text{ed}_k(G; p) \leq 1$ and $\text{ed}_k(G; q) = 0$ for $q \neq p$, so Theorem 10.10 only says $p \mid d_0$, consistent with $d_0 = p^n$; it says nothing about $d_1, \dots, d_{n-1}$, which is the point of this note. In characteristic 0 the constraint is much stronger.

Proposition 10.12 (q -groups). *Let G be a q -group of order q^n , $\text{char } k \neq q$ (k perfect if positive), $c = [k(\mu_q) : k]$. Then: (a) $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k(G; q)$ for $c \leq d \leq q - 1$, and $\text{ed}_k(G; q) \leq \text{ed}_k^{[d]}(\tau_{\text{gen}}) \leq \text{ed}_k(G)$ for $d < c$; (b) $\text{ed}_k(G; q) = \text{ed}_{k(\mu_q)}(G)$ is the minimal dimension of a faithful $k(\mu_q)$ -representation [26]; (c) $q \mid d_j(\tau_{\text{gen}})$ for every $j < \text{ed}_k(G; q)$, and $d_0 = q^n$; (d) if $\mu_q \subset k$: $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k(G)$ for $d < q$ and all steps of the profile occur at multiples of q , i.e. $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k^{[q \lfloor d/q \rfloor]}(\tau_{\text{gen}})$ for $d \geq q$. [proved, modulo [26] and $\text{ed}(\cdot; q)$ being invariant under prime-to- q extensions of k]*

Proof. Lower bounds: every degree $< q$ is prime to q (Theorem 10.10). Upper bound for $d \geq c$: Corollary 10.5 and $\text{ed}_{k(\mu_q)}(G) = \text{ed}_{k(\mu_q)}(G; q) = \text{ed}_k(G; q)$ [26, 29]. (c) is Theorem 10.10(b) and Proposition 10.2(c); (d) is (a)+(c) with $c = 1$. $\square$

Compare Proposition 3.7: in characteristic p the prime-to- p tail of a p -group collapses to ≤ 1 ; in characteristic $\neq q$ the prime-to- q tail of a q -group is *constant*, equal to $\text{ed}_k(G; q)$.

10.4. Geometric meaning and induced torsors in every characteristic.

Theorem 10.13. *Theorem 4.2 and Corollary 4.4 hold for every finite group G , every field k and every characteristic: $\text{ed}_k^{[d]}(\tau) \leq m$ iff some quasi-projective generically free primitive G -variety Z of dimension $\leq m$ has a closed point of degree $\leq d$ in the free locus of ${}^\tau Z$, iff there are such Z , F'/F of degree $\leq d$ and a G -equivariant morphism $T_{F'} \rightarrow Z^{\text{free}}$. For τ_{gen} this is a G -correspondence $V \rightsquigarrow Z$ of degree $\leq d$ into Z^{free} . [proved]*

Proof. The proof of Theorem 4.2 uses only: the existence of the twist ${}^\tau Z = (T \times_k Z)/G$ as a quasi-projective F -scheme compatible with base change and open immersions [21, §3]; the identification of F -points of ${}^\tau Z$ with equivariant maps $T \rightarrow Z_F$ (elementary for finite G); the fact that $Z^{\text{free}} \rightarrow Z^{\text{free}}/G$ is a G -torsor (SGA 1, V, finite group acting freely); and spreading out. None of these depends on the characteristic or on G being a cyclic p -group. $\square$

Lemma 10.14 (induced torsors). *For finite groups $H \leq G$ and an H -torsor τ_H over F , $\text{ed}_k(\text{Ind}_H^G \tau_H) = \text{ed}_k(\tau_H)$. [proved]*

Proof. “ $\leq$ ”: descend and induce. “ $\geq$ ”: reduce to T_H a field (if $T_H = \text{Ind}_{H'}^H T'$ with T' a field, then $\text{Ind}_H^G \tau_H = \text{Ind}_{H'}^G \tau'$ and $\text{ed}(\tau_H) \leq \text{ed}(\tau')$). Then the argument of Proposition 3.9 goes through for non-abelian G : a descent $T_0 \rightarrow \text{Spec } F_0$ has a connected component T'_0 with stabiliser H_0 ; the components of $T'_0 \otimes_{F_0} F$ are among the gT_H and are permuted transitively by H_0 , so after conjugating H we may assume $T_H \subseteq T'_0 \otimes_{F_0} F$ and $H \leq H_0$. With $F_1 = (T'_0)^H$ and $F_1 \otimes_{F_0} F = \prod F_i$, the factor $T'_0 \otimes_{F_1} F_{i_0}$ containing T_H is an H -torsor over F_{i_0} on whose components H acts transitively; being H -stable, T_H is that factor, and $F_{i_0} = T_H^H = F$. So τ_H descends to F_1 , $\text{trdeg}_k F_1 = \text{trdeg}_k F_0$. $\square$

Corollary 10.15 (subgroup profile). *For every subgroup $H' \leq G$: $\text{ed}_k^{[[G:H']]}(\tau_{\text{gen}}) \leq \text{ed}_k(\tau_{\text{gen}} \otimes L^{H'}) = \text{ed}_k(H')$. Hence $d_j(\tau_{\text{gen}}) \leq \min\{[G:H'] : \text{ed}_k(H') \leq j\}$. [proved]*

Proof. $\tau_{\text{gen}} \otimes L^{H'} = \text{Ind}_{H'}^G(L/L^{H'})$, and $L/L^{H'}$ is the generic H' -torsor of $V|_{H'}$ over k ; apply Lemma 10.14. $\square$

This is the general form of Proposition 3.9 and of the root-field reductions of Example 10.20.

10.5. Examples in characteristic zero.

Example 10.16 ($\mathbb{Z}/n$). Let $\mu_n \subset k$, $\text{char } k \nmid n$. The generic torsor $K(\sqrt[n]{a})/K$, $K = k(a)$, has $\text{ed} = 1$ and $d_0 = n$ (Proposition 10.2), so its profile is $(1, \dots, 1, 0)$. [proved]

Theorem 10.17 ($(\mathbb{Z}/2)^n$). *Let k be algebraically closed of characteristic $\neq 2$, $G = (\mathbb{Z}/2)^n$, $F = K = k(a_1, \dots, a_n)$, $\tau_{\text{gen}} = (a_1, \dots, a_n) \in H^1(F, \mathbb{Z}/2)^n$.*

- (a) $\text{ed}^{[2^j]}(\tau_{\text{gen}}) \leq n - j$ for $0 \leq j \leq n$, and $d_0 = 2^n$.
- (b) Every F'/F of odd degree has $\text{ed}(\tau_{\text{gen}} \otimes F') = n$; all d_j , $j < n$, are even.
- (c) For every quadratic F'/F , $\text{ed}(\tau_{\text{gen}} \otimes F') \geq n - 1$. Hence $\text{ed}^{[2]} = \text{ed}^{[3]} = n - 1$, $d_{n-1} = 2$, and $d_{n-2} = 4$ for $n \geq 2$.

- (d) $\text{ed}^{[d]} = 1$ for $2^{n-1} \leq d < 2^n$. Complete profiles: $(1, 0)$ for $n = 1$; $(2, 1, 1, 0)$ for $n = 2$; $(3, 2, 2, 1, 1, 1, 0)$ for $n = 3$.

[proved; inputs cited]

Proof. (a) Over $F(\sqrt{a_1}, \dots, \sqrt{a_j})$ the class is $(1, \dots, 1, a_{j+1}, \dots, a_n)$. (b) $\text{ed}_k(G; 2) = \text{ed}_k(G) = n$ [19, 26]; apply Theorem 10.10. (d) from (a), (b), (c) and Proposition 10.2.

(c) Write $(a_I) = \bigcup_{i \in I} (a_i) \in H^{|I|}(F, \mathbb{Z}/2)$. Suppose $\text{ed}(\tau_{\text{gen}} \otimes F') \leq n-2$ for $F' = F(\sqrt{f})$. The class descends to F_0 with $\text{trdeg}_k F_0 \leq n-2$, so $H^{n-1}(F_0, \mathbb{Z}/2) = 0$ [33, II.4.2] and $(a_I)_{F'} = 0$ for all $|I| = n-1$. By the Arason exact sequence $H^{r-1}(F) \xrightarrow{(f) \cup} H^r(F) \xrightarrow{\text{res}} H^r(F(\sqrt{f}))$ [17], $(a_{[n] \setminus i}) \in (f) \cup H^{n-2}(F)$ for all i . We show by induction on m that for $F_m = k(a_1, \dots, a_m)$ there is no $g \in F_m^\times$ with $(a_{[m] \setminus i}) \in (g) \cup H^{m-2}(F_m)$ for all $i \in [m]$. For $m = 1$: $(a_\emptyset) = 1 \neq 0$ in H^0 . For $m \geq 2$ pass to $\hat{F} = F_{m-1}((a_m))$, where $H^r(\hat{F}) = H^r(F_{m-1}) \oplus H^{r-1}(F_{m-1}) \cup (a_m)$ and units have $(u) = (\bar{u})$ [33, II.4.3]; the classes (a_I) restrict injectively (their residues are nonzero by induction on the residue maps). Write $g = a_m^\varepsilon u$ and the given $x_i = y_i + z_i \cup (a_m)$. If $\varepsilon = 1$, take $i = m$: comparing the two summands of $(a_{[m-1]}) = (a_m u) \cup x_m$ gives $y_m = (\bar{u}) \cup z_m$ and $(a_{[m-1]}) = (\bar{u}) \cup (\bar{u}) \cup z_m = 0$ (as $-1 \in k^{\times 2}$), a contradiction. If $\varepsilon = 0$, for $i < m$ the second summands give $(a_{[m-1] \setminus i}) = (\bar{u}) \cup z_i$ for all $i \in [m-1]$, contradicting the induction hypothesis with $g = \bar{u}$. So no quadratic extension lowers ed by two; with (a), (b) this gives $\text{ed}^{[2]} = \text{ed}^{[3]} = n-1$, $d_{n-1} = 2$, $d_{n-2} = 4$. $\square$

Remark 10.18. Whether $\text{ed}^{[2^j]}(\tau_{\text{gen}}) = n-j$ for $2 \leq j \leq n-2$ is open; the first case is $n = 4$: is there a quartic extension of $k(a_1, \dots, a_4)$ splitting all six quaternion algebras (a_i, a_j) ? Non-Galois quartics have no kernel description as clean as Arason's, which is what makes $\text{ed}^{[4]}$ hard. [open]

Example 10.19 ($\mathbb{Z}/p$ over $\mathbb{Q}$). Let p be an odd prime and τ any $\mathbb{Z}/p$ -torsor over $F \supseteq \mathbb{Q}$. By Corollary 10.5 and Example 10.16, $\text{ed}_{\mathbb{Q}}^{[F(\mu_p):F]}(\tau) \leq 1$, so $d_1(\tau) \leq p-1$ for every such torsor, although $\text{ed}_{\mathbb{Q}}(\mathbb{Z}/p) \geq 2$ for $p \geq 5$ [28]: $\text{ed}_k(G) = 1$ in characteristic 0 iff $G \hookrightarrow \text{PGL}_2(k)$, and $\mathbb{Z}/p \not\hookrightarrow \text{PGL}_2(\mathbb{Q})$ for $p \geq 5$ since $\zeta_p + \zeta_p^{-1} \notin \mathbb{Q}$. The profile of the generic $\mathbb{Z}/p$ -torsor over $\mathbb{Q}$ thus drops to 1 at a degree dividing $p-1$, prime to p , consistent with Theorem 10.10 since $\text{ed}_{\mathbb{Q}}(\mathbb{Z}/p; p) = 1$. Whether $d_1(\tau_{\text{gen}}) = [\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}] = (p-1)/2$ is open. [proved from [28]]

Example 10.20 (symmetric groups). Let $\text{char } k = 0$, $L = k(x_1, \dots, x_n)$, $F = K = k(s_1, \dots, s_n)$, so τ_{gen} is the generic degree- n étale algebra. For $F_m = F(x_1, \dots, x_m)$, $[F_m : F] = n(n-1) \cdots (n-m+1)$, Corollary 10.15 and (B4) give

$$\text{ed}_k(\tau_{\text{gen}} \otimes F_m) = \text{ed}_k(S_{n-m}), \quad \text{so} \quad \text{ed}_k^{[n(n-1) \cdots (n-m+1)]}(\tau_{\text{gen}}) \leq \text{ed}_k(S_{n-m}).$$

With $\text{ed}_k(S_n; q) = \lfloor n/q \rfloor$ [30], Theorem 10.10 gives $\text{ed}^{[d]}(\tau_{\text{gen}}) \geq \lfloor n/q_d \rfloor$ for the least prime $q_d > d$, so the profile of the generic étale algebra decays no faster than $\sim n/(2d)$, whereas the root-field bound needs degree $\sim n^m$ for a drop of m . For S_6 (with $\text{ed}_k(S_6) = 3$, $\text{ed}_k(S_5) = \text{ed}_k(S_4) = 2$, $\text{ed}_k(S_3) = 1$ [19], and $\text{ed}_k(A_6) = 3$ [cited from memory]): $\text{ed}^{[1]} = 3$; $\text{ed}^{[d]} \geq 2$ for $2 \leq d \leq 5$; $\text{ed}^{[6]} \leq 2$, $\text{ed}^{[120]} \leq 1$, $d_0 = 720$; $2 \mid d_2$, $6 \mid d_1$; so $d_2 \in \{2, 4, 6\}$ and $d_1 \in 6\mathbb{Z} \cap [6, 120]$. The discriminant field L^{A_6} is *not* a witness for $d_2 = 2$, since over it $\text{ed} = \text{ed}_k(A_6) = 3$. [proved from cited inputs]

10.6. Provenance: rewrites versus new statements.

Statement	Status relative to the literature
Prop. 10.1 (shape)	Rewrite of Prop. 3.2 for arbitrary G, k ; formal.
Prop. 10.2 (last jump)	New but elementary; (a) is folklore, (b) the regularity refinement is new.
Prop. 10.4, Cor. 10.5	New (elementary); the values quoted are [19, 26].
Lemma 10.7	Rewrite of [31, (2.5)]; the embedding argument is standard.
Lemma 10.8	Rewrite of Prop. 3.6 in a form valid for every finite G (no Witt vectors).
Prop. 10.9	Known: [31, 29] (ed_q of a versal object).
Thm. 10.10	The mechanism is [31, Thm. 10.1]; the statement about jump degrees and the bound (c) are new.
Prop. 10.12	New corollary of [26].
Thm. 10.13	Rewrite of Thm. 4.2 in full generality; ingredients [21, 31].
Lemma 10.14, Cor. 10.15	Extends Prop. 3.9 to non-abelian G ; probably known to experts, we found no reference.
Ex. 10.16	Folklore.
Thm. 10.17	(a),(b),(d) are immediate from the above; (c), i.e. $\text{ed}^{[2]} = \text{ed}^{[3]} = n - 1$ and $d_{n-2} = 4$, is new (Arason sequence [17] + residue induction).
Ex. 10.19	New observation from [28].
Ex. 10.20	The identity $\text{ed}(\tau_{\text{gen}} \otimes F_m) = \text{ed}(S_{n-m})$ is new but easy; inputs [19, 30].

10.7. Discussion and open questions. In characteristic 0 the profile of τ_{gen} is squeezed between Sylow divisibility (Theorem 10.10: a prime q with $\text{ed}_k(G; q) > j$ divides every $d_i, i \leq j$, and the prime-to- q tail is constant), the roots-of-unity layer (Corollary 10.5: within degree $[k(\mu_e) : k]$, prime to $|G|$ when $|G|$ is a prime power, the profile drops to the representation-theoretic value $\text{ed}_{k(\mu_e)}(G)$), and the subgroup upper bounds (Corollary 10.15). What is unknown is whether non-Galois or non-subgroup extensions can beat the subgroup witnesses. In characteristic p for a p -group (Sections 3–9) there is no Sylow divisibility beyond $p \mid d_0$ and no roots-of-unity layer; the whole profile lives in p -power degrees and records Artin–Schreier–Witt layers. Theorem 10.13 places both regimes in one framework: the profile is the minimal degree of a G -correspondence into a free G -variety of bounded dimension, Theorem 10.10 isolates the part of that degree forced by [31, Thm. 10.1], and the rest, the q -power part in characteristic $\neq q$ and the p -power part in characteristic p , is where the profile carries information beyond ed and $\text{ed}(\cdot; q)$.

Question 10.21. (1) For $(\mathbb{Z}/2)^n$ over an algebraically closed field of characteristic $\neq 2$, is $d_j(\tau_{\text{gen}}) = 2^{n-j}$ for all j (known for $j \in \{0, 1, n-1, n\}$)? For $(\mathbb{Z}/q)^n$ with $\mu_q \subset k$, the cyclic-kernel analogue of the Arason sequence should give $d_{n-1} = q$ [expected]. (2) For $\mathbb{Z}/p$ over $\mathbb{Q}$, is $d_1(\tau_{\text{gen}}) = (p-1)/2$? More generally, when is Corollary 10.5 sharp, and can an extension not of the form Fk' do better than adjoining roots of unity? (3) Determine $d_2(\tau_{\text{gen}}(S_6)) \in \{2, 4, 6\}$; does $d_j(\tau_{\text{gen}}(S_n))$ grow linearly or super-polynomially in n ? (4) Is the profile of $\tau_{\text{gen}}(V)$ independent of the faithful representation V ? (Presumably yes, by specialisation as in [31].)

11. SUPPLEMENTS AT $(p, n) = (2, 2)$ AND $(2, 3)$: THE CUBIC SLOT, QUADRATIC WITNESSES, AND A RIEMANN–ROCH BOUND

Addendum (27 September 2026). This section settles the open item of Theorem 6.2, gives a general Riemann–Roch bound for the prime-to- p jump degree, shows that the Artin–Schreier–Witt curve never carries a quadratic witness at $(2, 3)$ and that a quadratic witness can never have a descent field inside K , produces new families of quadratic extensions with essential dimension exactly 2, and reformulates the cubic slot. Notation is that of Sections 2, 5–7: k algebraically closed of characteristic 2, $K = k(s_0, s_1, s_2)$, $L = k(z_0, z_1, z_2)$, $s = \wp(z)$, $K_2 = k(s_0, s_1)$, τ_2 the generic $\mathbb{Z}/4$ -torsor, $E' : w^2 + w = u(u+1)(u+s_0) + s_1$, $\bar{U}_n$ the Artin–Schreier–Witt curve $\wp y = [t]$, Φ_2 the polynomial of Proposition 7.5 (written out in (4)). We write $(s_0, s_1)_K$ for the $\mathbb{Z}/4$ -class $(s_0, s_1) \in W_2(K)/\wp$, i.e. $\tau_2 \otimes_{K_2} K$; by Lemma 4.1(i), $\text{ed}_k((s_0, s_1)_K) = \text{ed}_k(\tau_2) = 2$. Every separable quadratic extension of K is $K(v_f)$, $v_f^2 + v_f = f$, $f \in K/\wp K$ (Proposition 3.6). Scripts referred to are listed in Appendix F.

11.1. Supplement to Theorem 6.2: the prime-to-2 jump degree at $n = 2$.

Theorem 11.1. *Let $p = 2$, $n = 2$. Let u be a root of*

$$P(u) := u^3 + (1 + s_0)u^2 + s_0u + s_1 = u(u+1)(u+s_0) + s_1.$$

Then $F' = K_2(u)$ is a separable cubic extension of K_2 and $(s_0, s_1) \equiv [s_0 + u^2 + u] \bmod \wp W_2(F')$. Hence $\text{ed}_k(\tau_2 \otimes F') = 1$ and

$$d_1^{(p')}(\tau_2) = 3.$$

More precisely, the cubic closed points of E' over K_2 are exactly the sections by the lines $w = \alpha u + \beta$, $\alpha, \beta \in K_2$; each is a single free point of degree 3, and over its residue field (s_0, s_1) is Teichmüller. Consequently $D(U_2) = \mathbb{Z}_{\geq 2}$, and the profile of τ_2 is completely known: $\text{ed}^{[1]} = 2$, $\text{ed}^{[2]} = \text{ed}^{[3]} = 1$, $\text{ed}^{[4]} = 0$, $d_1 = 2$, $d_1^{(p')} = 3$, $d_0 = 4$. [proved]

Proof. $P \in k(s_0)[u][s_1]$ is of degree 1 in s_1 with coprime coefficients 1 and $u(u+1)(u+s_0)$, hence irreducible in $k(s_0)[u, s_1]$, hence (Gauss, P monic in u) irreducible in $K_2[u]$; an irreducible cubic in characteristic 2 is separable. By (3)–(4), for $h = (u, 0) \in W_2(F')$: $(s + \wp h)_0 = s_0 + u^2 + u$ and $(s + \wp h)_1 = s_1 + u^3 + u^2 + s_0u^2 + s_0u = P(u) = 0$, so $s + \wp h = [s_0 + u^2 + u]$; equivalently $(u, 0)$ is an F' -point of $E'^\circ = \tau_2 U_2$ and the last sentence of Theorem 6.2 applies. So $\text{ed}_k(\tau_2 \otimes F') \leq 1$, and ≥ 1 since $3 < 4 = d_0$. As $d_1^{(p')}$ is odd and $\geq d_1(\tau_2) = 2$ (Proposition 6.1), $d_1^{(p')}(\tau_2) = 3$.

Cubic points: $E'(K_2) = \{\infty'\}$ (Theorem 6.2), so $\text{Pic}^0(E')(K_2) = 0$ and every effective K_2 -divisor of degree 3 is linearly equivalent to $3\infty'$, i.e. is the zero divisor of an element of $L(3\infty') = \langle 1, u, w \rangle$; the sections $u = c$ are a degree-2 fibre plus ∞' , so the cubic points are the sections by lines $w = \alpha u + \beta$. Such a section is a single closed point of degree 3 because a K_2 -rational component would be a K_2 -point of E'° . It lies in E'° , hence is free. $D(U_2) \supseteq \{2, 3\}$ and Appendix B gives all of $\mathbb{Z}_{\geq 2}$. $\square$

Remark 11.2. (i) This closes the item marked [open] in Theorem 6.2 and the $p = 2$ part of Question 6.4: the “group-theoretic” obstruction $E'(K_2) = 0$ kills degree-1 points and produces the cubic ones. (ii) For odd p the same mechanism gives $d_1^{(p')}(\tau_2) \leq p^2 - p + 1$ (Theorem 11.3); Question 6.4 for degrees in $[2, p - 1]$ is untouched.

11.2. Supplement to Proposition 3.7 and Remark 4.5: a Riemann–Roch bound.

Let $l_n = l_n(p)$ be the largest lower ramification break of $\bar{U}_n \rightarrow \mathbb{P}_t^1$ at ∞ , i.e. the pole order at

∞ of the reduced top Witt coordinate of U_n ; by Appendix B,

$$(1) \quad l_n(p) = 1 + (p-1) \sum_{j=1}^{n-1} p^{2j-1} : \quad l_2(2) = 3, \quad l_2(p) = p^2 - p + 1, \quad l_3(2) = 11, \quad l_4(2) = 43.$$

Theorem 11.3. *For every (p, n) , $D(U_n)$ contains an integer prime to p and $\leq l_n$; hence*

$$d_1^{(p')}(\tau_{\text{gen}}) \leq l_n(p).$$

This is the best bound obtainable from the pencils $|d \cdot \infty''|$ on $\tau_{\text{gen}} \bar{U}_n$: every function regular on the affine curve U_n with pole order prime to p has pole order $\geq l_n$. [proved; the closed formula (1) is proved for $(2, 2)$, $(2, 3)$, $(p, 2)$ and cited in general]

Proof. $X := \tau_{\text{gen}} \bar{U}_n$ is a smooth projective geometrically integral K -curve with the K -point ∞'' (twist of the G -fixed point), and its free locus is $X \setminus \{\infty''\}$ (Proposition 3.7(c)). Riemann–Roch dimensions $\ell(d\infty'')$ are invariant under base change to $\bar{K}$, where $(X, \infty'') \cong (\bar{U}_n, \infty)$; so the Weierstrass semigroup of ∞'' on X is that of ∞ on $\bar{U}_n$. The top layer $U_n \rightarrow U_{n-1}$ is $y^p - y = c$ with $v_\infty(c) = -l_n$ on U_{n-1} and ∞ totally ramified, so $v(y) = -l_n$ on U_n and l_n is a pole order. Hence there is $f \in K(X)$ with pole divisor exactly $l_n \infty''$. For $c \in K$ the divisor $\text{div}(f - c) + l_n \infty''$ is effective, K -rational, of degree l_n and supported on $X \setminus \{\infty''\}$. Its closed points have degrees summing to $l_n \equiv 1 \pmod{p}$, so one of them has degree e prime to p , $e \leq l_n$, and it is free; Theorem 4.2 gives $\text{ed}_k^{[e]}(\tau_{\text{gen}}) \leq 1$. For optimality: the coordinate ring of U_n is free over that of U_{n-1} with basis $1, y, \dots, y^{p-1}$, and $v(a_j y^j) = p v_{U_{n-1}}(a_j) - j l_n$ are pairwise distinct modulo p , so a function with pole order prime to p has a dominant term with $j \geq 1$ and pole order $\geq l_n$. $\square$

Corollary 11.4. *At $(p, n) = (2, 3)$: the fibres of the pole-order-11 pencil are free divisors of degree 11 without degree-1 components (Proposition 7.5), so each contains a free point of odd degree in $\{3, 5, 7, 9, 11\}$, and $11 \in D(U_3)$ exactly (Appendix B). Hence*

$$d_1^{(p')}(\tau_{\text{gen}}) \in \{3, 5, 7, 9, 11\}, \quad \delta_{\text{free}}(U_3) \in \{3, 5, 7, 9, 11\},$$

the second by Corollary 11.7 below. This refines “ $d_1^{(p')} \geq 3$ finite” in Theorem 7.1. [proved]

Remark 11.5. The naive argument “ $\ell(d\infty'') \geq 2$ for $d \geq g + 1$, take d odd” does not work: the fibres of $f \in L(d\infty'')$ have degree equal to the actual pole order of f , which lies in the Weierstrass semigroup and may be even. At $(2, 3)$, $g(\bar{U}_3) = 7$ and the semigroup of ∞ is $\langle 4, 6, 11 \rangle$, whose gaps are $\{1, 2, 3, 5, 7, 9, 13\}$: every function of pole order ≤ 10 has even pole order.

11.3. Supplement to Section 7: quadratic witnesses. Fix $f \in K \setminus \wp K$ and $K' = K(v_f)$. A *witness* is such a K' with $\text{ed}_k(\tau_{\text{gen}} \otimes K') = 1$; a *descent field* is a subfield $k \subset F_0 \subset K'$ of transcendence degree 1 with a class $\beta \in W_3(F_0)/\wp$, $s \equiv \beta$ in $W_3(K')/\wp$.

Theorem 11.6 (no descent field inside K). *Let $K' = K(v_f)$ be a witness with descent field F_0 and class β .*

- (a) *If $F_0 \subset K$ then $F_0 = k(t)$ is rational, $s \equiv \beta + \varepsilon(0, 0, f) \pmod{\wp W_3(K)}$ with $\varepsilon \in \{0, 1\}$, and $\text{ed}_k(\tau_{\text{gen}}) \leq \text{trdeg}_k k(t, f) \leq 2$.*
- (b) *The case (a) does not occur: $F_0 \not\subset K$.*
- (c) *The truncation of β is a descent of $(s_0, s_1)_K \otimes K'$, so $f \in \mathfrak{F} := \{f \in K/\wp K : \text{ed}_k((s_0, s_1)_K \otimes K(v_f)) = 1\}$.*

[proved]

Proof. (a) $\ker(H^1(K, \mathbb{Z}/8) \rightarrow H^1(K', \mathbb{Z}/8)) = \text{Hom}(\mathbb{Z}/2, \mathbb{Z}/8) \cong \mathbb{Z}/2$ by inflation–restriction, generated by the induced class $V^2[f] = (0, 0, f)$ (Section 2.2). Since s and β both lie in $W_3(K)/\wp$ and agree over K' , $s \equiv \beta + \varepsilon(0, 0, f) = (\beta_0, \beta_1, \beta_2 + \varepsilon f)$ by (2); and a subfield of transcendence degree 1 of $k(s_0, s_1, s_2)$ is $k(t)$ (a curve dominated by $\mathbb{A}^3$ is dominated by a line, Lemma 2.5(i)). (b) Truncate the congruence of (a) to W_2 : the term $(0, 0, f)$ dies and $(s_0, s_1) \equiv (\beta_0, \beta_1) \pmod{\wp W_2(K)}$ with $\beta_0, \beta_1 \in F_0$, so $\text{ed}_k((s_0, s_1)_K) \leq 1$, contradicting $\text{ed}_k((s_0, s_1)_K) = 2$. (c) Truncation commutes with $\wp$ and with base change. $\square$

So a quadratic witness, if one exists, has a descent field *not* contained in K and can never be turned into a counterexample to Ledet’s conjecture by this route; the truncation argument is the one used in the proof of Proposition 7.5 (the step “ $\varepsilon = 1 \Rightarrow$ a K -point of E' ”), applied to an arbitrary descent class instead of $[t]$.

Corollary 11.7 ($\bar{U}_3$ admits no quadratic witness). *For every quadratic K'/K , $\tau_{\text{gen}}\bar{U}_3$ has no free K' -point: s is not Teichmüller over any quadratic extension of K , and $D(U_3) \cap \{1, 2\} = \emptyset$. In particular Algorithm 9.1 has nothing to find, and the “points not coming from the s_2 -line” of Remark 7.6 do not exist.* [proved]

Proof. By Proposition 7.5 a free K' -point needs $Q = (u, w_1) \in E'(K')$ with $u \notin K$. Let σ generate $\text{Gal}(K'/K)$. $Q + \sigma Q$ is σ -invariant, hence lies in $E'(K) = \{\infty'\}$ (Lemma 4.1(ii)); so $\sigma Q = -Q = (u, w_1 + 1)$, whence $\sigma(u) = u$ and $u \in K$. $\square$

The same argument at level 2 shows that $E'(K(v_f)) \neq \{\infty'\}$ iff $f \equiv s_1 + u(u+1)(u+s_0) \pmod{\wp K}$ for some $u \in K$: the E_0 -witnesses of Theorem 6.2 are exactly the u -fibres (Proposition D.2).

Proposition 11.8 (fixed-field trick). *Let $K' = K(v_f)$ with $f \notin \{0, s_0\} + \wp K$, so that $L' = L \otimes_K K' = L(v_f)$ is a field. Suppose there are a subfield $M \subset L'$ and $x \in L'$ with $L' = M(x)$ purely transcendental, such that M is fixed pointwise by some $1 \neq h \in G$. Then $\text{ed}_k(\tau_{\text{gen}} \otimes K') \geq 2$. The same holds for $(s_0, s_1)_K \otimes K'$, with L' replaced by $k(z_0, z_1, s_2)(v_f)$ and G by $\mathbb{Z}/4$.* [proved]

Proof. A witness gives, by Corollary 4.4, a smooth projective curve C with faithful G -action and a G -equivariant dominant rational map from a model of L' ; take the model $B \times \mathbb{A}^1$ with $k(B) = M$. Lemma 7.2 for $B \times \mathbb{A}^1 \rightarrow B$ gives: either a fibre $\mathbb{A}^1$ maps nonconstantly to C , so $C \cong \mathbb{P}^1$ (Lemma 2.5(i)), impossible (Lemma 2.5(ii)); or $k(C) \subset M \subset (L')^h$, so h acts trivially on C , contradicting faithfulness. $\square$

Theorem 11.9 (quadratic extensions with ed exactly 2). $\text{ed}_k(\tau_{\text{gen}} \otimes K(v_f)) = 2$ for every $f \notin \wp K$ in each of the classes

- (a) $f \in K_2 + \wp K$ (this contains $K(v_1)$ and $K(z_0)$ of Theorem 7.3);
- (b) $f \in s_2 + K_2 + \wp K$ (this contains $K(v_2)$, not covered by Theorem 7.3);
- (c) $f \in k(g) + \wp K$ for any $g \in K$ with $K = K_2(g)$, e.g. $f \in k(s_2)$, $f = 1/(s_2 + s_0 s_1)$;
- (d) $f \in k(s_0, s_1)[s_2]$ a polynomial in s_2 of degree ≤ 7 .

In cases (a), (c) already $\text{ed}_k((s_0, s_1)_K \otimes K(v_f)) = 2$, i.e. $f \notin \mathfrak{F}$. [proved]

Proof. The upper bound is Theorem 7.1. (a) Replace f by $e \in K_2$ with $f \equiv e$. If $e \equiv s_0$, Proposition 3.9. Otherwise $M := k(z_0, z_1, v_e)$ is algebraic over $k(z_0, z_1)$, $L' = M(z_2)$, and $g^4: z \mapsto z + V^2(1) = (z_0, z_1, z_2 + 1)$ fixes M ; apply Proposition 11.8. (b) Write $f = s_2 + e + \wp h$, $e \in K_2$. Over $K' = K(v_f)$, $s_2 = \wp(v_f + h) + e$, so $s = (s_0, s_1, e) + \wp V^2(v_f + h) \equiv (s_0, s_1, e)$ by (2). Now $K' = K_2(v_f)$ is purely transcendental over K_2 , so by Lemma 4.1(i) $\text{ed}_k(\tau_{\text{gen}} \otimes K') = \text{ed}_k((s_0, s_1, e)_{K_2}) \geq \text{ed}_k((s_0, s_1)_{K_2}) = 2$. (c) Replace f by an element of $k(g)$. Put $F := k(g, v_f)$; then $K' = F(s_0, s_1)$ with s_0, s_1 algebraically independent over F , and the total space of $(s_0, s_1)_K \otimes K'$ is $k(z_0, z_1, g, v_f) = F(z_0, z_1)$, a field on which $\mathbb{Z}/4$ acts fixing F ; the

argument of Proposition 11.8 with the fibration $B \times \mathbb{A}^2 \rightarrow B$ excludes a faithful $\mathbb{Z}/4$ -curve, so $\text{ed}_k((s_0, s_1)_K \otimes K') = 2$, and a descent of s over K' would truncate to a descent of $(s_0, s_1)_K \otimes K'$. (d) If $f \in K_2 + \wp K$ use (a); otherwise Theorem 11.10(b) gives a witness curve of p -rank 0 and genus ≤ 6 , contradicting Corollary 11.11. $\square$

Theorem 11.10 (structure of a quadratic witness). *Let $K' = K(v_f)$ be a witness with $f \notin K_2 + \wp K$, C its curve, $C' = C/G$. For $a \in k^2$ let $D_{f,a}: y^2 + y = f(a_0, a_1, s_2)$ (a curve over k in the s_2 -line) and, for $b \in \wp_2^{-1}(a) \subset \mathbb{A}_z^2$, let D_b be its pullback along $z_2 \mapsto s_2 = z_2^2 + z_2 + C(b_0, b_1)$.*

- (a) *For a in a dense open subset of $\mathbb{A}^2(k)$: $D_{f,a}$ is integral and dominates C' , and every irreducible component of D_b dominates C . Hence $g(C) \leq g(\hat{D}_b)$ and $\sigma(C) \leq \sigma(\hat{D}_b)$ (σ the p -rank, $\hat{D}_b$ the normalisation of a component).*
- (b) *If $f \in k(s_0, s_1)[s_2]$ has s_2 -degree $m \geq 1$, then $\sigma(C) = 0$, $C \rightarrow C'$ is totally ramified at exactly one point and étale elsewhere, C' has p -rank 0 and genus $\leq \lfloor (m-1)/2 \rfloor$, and $7 \leq g(C) \leq m-1$. In particular $m \geq 8$, and for $8 \leq m \leq 15$ one has $C' \cong \mathbb{P}^1$: C is the Artin–Schreier–Witt curve of a polynomial Witt vector $\beta \in W_3(k[t])$, and $s \equiv \beta(t)$ over K' with $t \in K' \setminus K$.*

[proved]

Proof. (a) Let $B_f: y^2 + y = f(s_0, s_1, s_2)$ over $\mathbb{A}_s^3$, $k(B_f) = K'$. Its generic fibre over $\mathbb{A}_{(s_0, s_1)}^2$ is the Artin–Schreier curve $y^2 + y = f$ over K_2 in the s_2 -line, geometrically integral because $f \notin K_2 + \wp K$ (a constant quadratic subextension $K_2(v_e) \subset K'$ would force $f \equiv e$). Apply Lemma 7.2 to $B_f \rightarrow \mathbb{A}_s^3$ and the dominant rational map $B_f \dashrightarrow C'$ given by $k(C') \subset K'$: the alternative $k(C') \subset k(s_0, s_1) \subset K$ is excluded by Theorem 11.6(b), so $D_{f,a} \dashrightarrow C'$ is dominant for general a . Let Y_f be the normalisation of $\mathbb{A}_z^3 \times_{\mathbb{A}_s^3} B_f$, with function field L' ; the equivariant map $Y_f \dashrightarrow C$ composed with $C \rightarrow C'$ factors through $Y_f \rightarrow B_f \dashrightarrow C'$, and the fibre of Y_f over a is $\bigsqcup_{b \in \wp_2^{-1}(a)} D_b$, permuted transitively by G . Each $D_b \rightarrow D_{f,a} \rightarrow C'$ is nonconstant, hence so is $D_b \rightarrow C$. Genus and p -rank do not increase under dominant maps of curves. (b) $D_{f,a}$ is $y^2 + y = (\text{polynomial of degree } m \text{ in } s_2)$, with a single branch point, so $\sigma(D_{f,a}) = 0$ (Deuring–Shafarevich) and $g \leq \lfloor (m-1)/2 \rfloor$; D_b is $y^2 + y = (\text{polynomial of degree } 2m \text{ in } z_2)$, again with one branch point, so $\sigma = 0$ and, after Artin–Schreier reduction to odd degree $\leq 2m-1$, $g(D_b) \leq m-1$. By (a), $\sigma(C) = \sigma(C') = 0$ and $g(C) \leq m-1$; Lemma C.1 and Corollary 11.11 give the ramification structure and $g(C) \geq 7$. If $g(C') = 1$ then Riemann–Hurwitz gives $2g(C) - 2 \geq 0 + 28$, so $m \geq 16$; for $m \leq 15$, $C' = \mathbb{P}^1$ and the cover is étale outside one point, so its class lies in $H^1(\mathbb{A}^1, \mathbb{Z}/8) = W_3(k[t])/\wp$, with β_0 nonconstant since $s_0 \notin \wp K'$ (else $K' = K(z_0)$). $\square$

Corollary 11.11 (minimal genus). *A smooth projective curve with faithful $\mathbb{Z}/8$ -action and p -rank 0 in characteristic 2 has genus ≥ 7 , with equality iff $C/G \cong \mathbb{P}^1$ and the upper ramification breaks at the unique ramified point are $(1, 2, 4)$; $\bar{U}_3$ is an example, $g(\bar{U}_3) = 7$. A faithful $\mathbb{Z}/4$ -curve of p -rank 0 has genus ≥ 1 , with equality iff it is $(E_0, \text{Witt action})$ of Theorem 6.2. [proved]*

Proof. Lemma C.1: one totally ramified point, C/G of p -rank 0. The different exponent at that point is $(u_1 + 1) + 2(u_2 + 1) + 4(u_3 + 1)$ in terms of the upper breaks, with $u_1 \geq 1$ and $u_{i+1} \geq 2u_i$ [23, 34], so $2g - 2 \geq -16 + 28 = 12$. For $\mathbb{Z}/4$: $2g - 2 \geq -8 + 2 + 6 = 0$, and the genus-1 case is (E_0, Witt) by Lemma C.2. $\square$

Remark 11.12. (i) Theorem 7.3's proofs for $K(v_1)$ and $K(v_1, v_2)$ are instances of the mechanism of Theorem 11.10; for $f \in K_2$ the fixed-field trick is shorter, but for the quartic tower it does not apply directly (g^4 moves $y' = v_2 + z_2$) and the argument of Theorem 7.3 is needed. (ii)

The only genus-1 faithful $\mathbb{Z}/8$ -curves are ordinary elliptic curves with G acting by translation by a point of order 8 (Appendix C); by Theorem 11.10(a) they can occur as witness curves only if f has, for general a , at least two poles on the s_2 -line, and then $\tau_{\text{gen}} \otimes K'$ is the étale $\mathbb{Z}/8$ -isogeny torsor of an ordinary elliptic curve at a non-constant point (Appendix D). This and polynomial f of s_2 -degree ≥ 8 are the only remaining shapes of a quadratic witness.

11.4. Supplement to Proposition 7.5 and Question (Q2): the cubic slot. Let F'/K be cubic (separable without loss, Proposition 3.6). Then $L \otimes_K F'$ is a field and Corollary 4.4 applies: a witness for $\text{ed}^{[3]}(\tau_{\text{gen}}) = 1$ is a faithful $\mathbb{Z}/8$ -curve C with a free F' -point on $\tau_{\text{gen}}C$.

Proposition 11.13. *A cubic witness curve C has genus ≥ 3 and, if its p -rank is 0, genus ≥ 7 . Ordinary elliptic curves with translation action carry no free point of odd degree.* [proved]

Proof. $\mathbb{P}^1$ and supersingular elliptic curves are excluded by Lemma 2.5. If C is an ordinary elliptic curve with G acting by translation by P of order 8 (the only genus-1 case, Appendix C), then $\tau_{\text{gen}}C$ is a principal homogeneous space under C whose period divides 8; period and index have the same prime divisors [27], so a nontrivial such torsor has no closed point of odd degree, and a trivial one has a K -point, free since the action is free, forcing $\text{ed}_k(\tau_{\text{gen}}) \leq 1$, false. Genus 2: the hyperelliptic involution is central, and the image of $\mathbb{Z}/8$ in $\text{Aut}(\mathbb{P}^1)$ is $\mathbb{Z}/8$ or $\mathbb{Z}/4$, excluded by Lemma 2.5(ii). p -rank 0: Corollary 11.11. $\square$

On $\bar{U}_3$ the question is completely explicit. By Proposition 5.2(a) and (3)–(4), a free F' -point of $\tau_{\text{gen}}\bar{U}_3$ is $h = (u, w_1, w_2) \in W_3(F')$ with

$$(2) \quad w_1^2 + w_1 = u(u+1)(u+s_0) + s_1, \quad w_2^2 + w_2 = e_Q := s_2 + \Phi_2(s_0, s_1, u, w_1),$$

and then $s \equiv [s_0 + u^2 + u]$ over F' .

Proposition 11.14 (reformulation). $\text{ed}^{[3]}(\tau_{\text{gen}}) = 1$ is witnessed on $\bar{U}_3$ iff there are $\alpha, \beta \in K$ such that, for u a root of

$$u^3 + (1 + s_0 + \alpha^2)u^2 + (s_0 + \alpha)u + (s_1 + \beta^2 + \beta) = 0,$$

one has $e_Q = s_2 + \Phi_2(s_0, s_1, u, \alpha u + \beta) \in \wp K(u)$. Every such cubic is irreducible over K , and these are all the cubic points of E' over K . [proved]

Proof. As in Theorem 11.1, $E'(K) = \{\infty'\}$ (Lemma 4.1(ii)) gives $\text{Pic}^0(E')(K) = 0$, so the cubic points of E' over K are the line sections, each irreducible. A free F' -point with $[F' : K] = 3$ has $F' = K(u)$: if $u \in K$ then $t = s_0 + u^2 + u \in K$ and $s \equiv [t]$ over F' would give $s \equiv [t]$ over K , since $\text{res}_{F'/K}$ is injective on $H^1(\cdot, \mathbb{Z}/8)$ ($\text{cores} \circ \text{res} = 3$ is a unit mod 8). $\square$

Proposition 11.15 (necessary condition and a partial obstruction). (a) *If $s \equiv [t]$ over a cubic $F' = K(u)$ then $s \equiv 3 \cdot \text{cores}_{F'/K}[t]$ in $W_3(K)/\wp$, where cores is the Witt-vector trace $[t] + [t'] + [t''] = (e_1, e_2, e_1e_3 + e_1^2e_2)$ (e_i the elementary symmetric functions of the conjugates). Coordinates 0 and 1 of this condition hold automatically for every line, and coordinate 2 reads*

$$(3) \quad \text{Tr}_{K(u)/K}(e_Q) = s_2 + R(s_0, s_1, \alpha, \beta) \in \wp K$$

with R the explicit polynomial of Appendix E.

- (b) (3) fails for all $\alpha, \beta \in K_2$, and more generally for all α, β with pole order ≤ 1 at $s_2 = \infty$. Hence a cubic witness on $\bar{U}_3$, if any, has a line with α or β of pole order ≥ 2 at $s_2 = \infty$.

[(a) proved, coordinates checked symbolically; (b) proved, with computer-verified coefficients]

Proof. (a) $\text{cores} \circ \text{res} = 3$ on $H^1(K, \mathbb{Z}/8)$ and $3^2 \equiv 1 \pmod{8}$. Corestriction is a morphism of δ -functors for $0 \rightarrow \mathbb{Z}/8 \rightarrow W_3 \xrightarrow{\mathcal{S}} W_3 \rightarrow 0$, and on H^0 it is the norm, i.e. the Witt trace; the formula for $[t] + [t'] + [t'']$ follows from (3), and the coordinate computations are in Appendix E. (b) For $\alpha, \beta \in K_2$, $s_2 + R$ has a simple pole at $s_2 = \infty$ (Remark 7.4). In general write $\alpha = \sum_{i \leq 1} a_i s_2^i$, $\beta = \sum_{i \leq 1} b_i s_2^i$ in $K_2((1/s_2))$. Membership in $\wp$ of the completion forces, for every odd $i_0 \geq 1$, the chain condition $\sum_j c_{i_0 2^j}^{2^{J-j}} = 0$ on the coefficients c_i of s_2^i in $s_2 + R$ (Appendix E). The only monomial of R producing s_2^{11} from exponents ≤ 1 is α^{11} , giving $c_{11} = a_1^{11}$ and $a_1 = 0$; then $c_3 = b_1^3$ gives $b_1 = 0$; then α, β are integral at ∞ and $s_2 + R$ has a simple pole, contradicting the chain condition for $i_0 = 1$. $\square$

Exhaustive searches over $\mathbb{F}_2$ -polynomial lines (about $2.4 \cdot 10^6$ pairs (α, β) of small degree, Appendix E) found no line satisfying (3). Note that (3) is only necessary: the kernel of cores on $H^1(K(u), \mathbb{Z}/2)$ is large. Our expectation is that no cubic witness exists on $\bar{U}_3$; whether another faithful $\mathbb{Z}/8$ -curve of genus ≥ 3 carries one is a separate question, so (Q2) remains open.

11.5. Updated status at (2, 3) and revised questions.

Theorem 11.16 (status at (2, 3)).

$$\text{ed}^{[1]} \in \{2, 3\}, \quad \text{ed}^{[2]} \in \{1, 2\}, \quad \text{ed}^{[3]} \in \{1, 2\}, \quad \text{ed}^{[4]} = \dots = \text{ed}^{[7]} = 1, \quad \text{ed}^{[8]} = 0;$$

$d_2 \in \{1, 2\}$, $d_1 \in \{2, 3, 4\}$, $d_0 = 8$, $d_1^{(p')} \in \{3, 5, 7, 9, 11\}$. Moreover: a quadratic witness for $d_1 = 2$ has no descent field inside K , is not $\bar{U}_3$, is not any f of Theorem 11.9, and for polynomial f is an Artin–Schreier–Witt curve of a polynomial Witt vector of s_2 -degree ≥ 8 ; $D(U_3) \cap \{1, 2\} = \emptyset$, $D(U_3) \supseteq \langle 4, 6, 11 \rangle \setminus \{0\}$, and $D(U_3) \cap \{3, 5, 7, 9, 13\}$ is unknown. [proved]

The open questions of Section 9.2 are refined as follows. (Q1): is there $f \in K/\wp K$, with at least two poles on the s_2 -line for general (s_0, s_1) or a polynomial of s_2 -degree ≥ 8 , such that $K(v_f)$ is a witness? The first shape requires the étale $\mathbb{Z}/8$ -isogeny torsor of an ordinary elliptic curve (Appendix D); the level-2 version (is $(s_0, s_1)_K \otimes K(v_f)$ the étale $\mathbb{Z}/4$ -isogeny torsor of an ordinary elliptic curve at a non-constant point?) is the natural first step, since a negative answer there kills all such level-3 witnesses by truncation. (Q2): does (3) fail for all $\alpha, \beta \in K$ (a local analysis at $s_2 = \infty$ for pole orders ≥ 2 is the missing piece), and which of 3, 5, 7, 9 lie in $D(U_3)$? The splitting type over K of the degree-11 fibres of the pole-order-11 pencil is a finite computation that would decide the latter. (Q3), (Q4) are unchanged, except that $d_1^{(p')}(\tau_2) \leq p^2 - p + 1$ for all p and $= 3$ for $p = 2$.

11.6. A quantitative Reichstein–Vistoli theorem. Reichstein–Vistoli [32] prove $\text{ed}_{k,p}(G) \leq 1$ for every finite p -group G in characteristic p , i.e. every G -torsor becomes one-dimensional over *some* extension of degree prime to p : in the language of the profile, $d_1^{(p')}(\tau) < \infty$ for every τ . The proof passes through the p -closure, a large field, and gives neither a bound on the degree nor uniformity in τ . The Riemann–Roch argument of Theorem 11.3 gives both. Throughout this subsection k is a field of characteristic p , algebraically closed unless said otherwise, G a finite group, and V a faithful representation of G over k with generic torsor τ_{gen} over $K = k(V)^G$.

Proposition 11.17 (the generic torsor dominates every profile). *Let τ be a G -torsor over an infinite field $F \supseteq k$. Then for every $d \geq 1$*

$$\text{ed}_k^{[d]}(\tau) \leq \text{ed}_k^{[d]}(\tau_{\text{gen}}), \quad \text{ed}_k^{[d]'}(\tau) \leq \text{ed}_k^{[d]'}(\tau_{\text{gen}})$$

(the second for the prime-to- p profiles), hence $d_j(\tau) \leq d_j(\tau_{\text{gen}})$ and $d_j^{(p')}(\tau) \leq d_j^{(p')}(\tau_{\text{gen}})$ for all j . Consequently $\sup_{\tau} \text{ed}_k^{[d]}(\tau) = \text{ed}_k^{[d]}(\tau_{\text{gen}})$, and the profile of τ_{gen} does not depend on the faithful representation V . [proved]

Proof. Let F'/K realise $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = m$, $[F' : K] = d' \leq d$; by Proposition 3.6 (or Lemma 10.8) we may take F'/K separable. By Theorem 10.13 there are a quasi-projective generically free primitive G -variety Z of dimension $\leq m$ and a G -equivariant morphism $\text{Spec}(L \otimes_K F') \rightarrow Z^{\text{free}}$. As in the proof of Theorem 4.2, let W' be the normalisation of V in $L \otimes_K F'$; then $\pi : W' \rightarrow V$ is finite, G -equivariant, generically étale of degree d' , and the equivariant map $W' \dashrightarrow Z^{\text{free}}$ is defined on a dense G -stable open U . Choose a dense G -stable open $V_1 \subseteq V$ over which π is étale and $\pi^{-1}(V_1) \subseteq U$. Twisting by τ (functoriality of twists, [21, §3]) gives a finite étale F -morphism ${}^{\tau}\pi^{-1}(V_1) \rightarrow {}^{\tau}V_1$ of degree d' and an F -morphism ${}^{\tau}\pi^{-1}(V_1) \rightarrow {}^{\tau}Z^{\text{free}}$. Now ${}^{\tau}V$ is an F -form of the vector space V_F , hence isomorphic to affine space by Hilbert 90, so ${}^{\tau}V_1(F)$ is non-empty as F is infinite; pick $x \in {}^{\tau}V_1(F)$. Its fibre is $\text{Spec } A$ for an étale F -algebra $A = \prod_i F_i$ of dimension d' , and each $\text{Spec } F_i \rightarrow {}^{\tau}Z^{\text{free}}$ gives a closed point of ${}^{\tau}Z$ in the free locus of degree $\leq [F_i : F]$. The degrees $[F_i : F]$ sum to $d' \leq d$, so $\text{ed}_k^{[d]}(\tau) \leq m$ by Theorem 10.13; if $p \nmid d'$, some $[F_i : F]$ is prime to p . The last two statements: the supremum is attained at τ_{gen} , and two generic torsors (for V and V') are torsors over infinite fields, so each dominates the other. $\square$

Definition 11.18. For a smooth projective curve C over k with a faithful G -action and a G -fixed k -point ∞ , let $\lambda(C, \infty)$ be the smallest element of the Weierstrass semigroup $H(\infty)$ that is prime to p , and let $\lambda_k(G)$ be the minimum of $\lambda(C, \infty)$ over all such pairs ($\lambda_k(G) = \infty$ if none exists).

Such pairs exist for every finite p -group and, more generally, for every $G \cong P \rtimes \mathbb{Z}/m$ with P a p -group and $p \nmid m$: by Harbater and Katz–Gabber [24, 25] there is a connected G -cover of $\mathbb{P}^1$ étale over $\mathbb{G}_m$ (over $\mathbb{A}^1$ if G is a p -group) with inertia group G at ∞ , and the point above ∞ is then G -fixed. Conversely a G -fixed point forces G to be an inertia group in characteristic p , hence of this form. For $G = \mathbb{Z}/p^n$ the pair $(\bar{U}_n, \infty)$ is defined over $\mathbb{F}_p$.

Theorem 11.19 (quantitative Reichstein–Vistoli). (a) *Let (C, ∞) be as in the definition. For every G -torsor τ over every field $F \supseteq k$ there is a separable extension F'/F with $p \nmid [F' : F] \leq \lambda(C, \infty)$ and $\text{ed}_k(\tau \otimes F') \leq 1$. Hence*

$$d_1^{(p')}(\tau) \leq d_1^{(p')}(\tau_{\text{gen}}) \leq \lambda_k(G) \quad \text{for every } \tau.$$

(b) *For $G = \mathbb{Z}/p^n$: $\lambda_k(\mathbb{Z}/p^n) \leq l_n(p) = 1 + (p-1) \sum_{j=1}^{n-1} p^{2j-1}$, realised by $(\bar{U}_n, \infty)$, and $\lambda(C, \infty) = l_n(p)$ is the minimum of $\lambda(C, \infty)$ over all connected $\mathbb{Z}/p^n$ -covers $C \rightarrow \mathbb{P}^1$ étale over $\mathbb{A}^1$. So every $\mathbb{Z}/p^n$ -torsor over every $F \supseteq k$ becomes one-dimensional over a prime-to- p extension of degree at most $l_n(p)$; the bound is 1 for $n = 1$, $p^2 - p + 1$ for $n = 2$, 11 at $(2, 3)$, 43 at $(2, 4)$, and it is attained by τ_{gen} for $p^n = 4$ (Theorem 11.1). Together with Proposition 3.4, $d_1(\tau) \leq \min(p^{n-1}, l_n(p))$ for every $\mathbb{Z}/p^n$ -torsor.*

(c) *For an arbitrary finite G with Sylow p -subgroup G_p : $d_1^{(p')}(\tau) \leq [G : G_p] \cdot \lambda_k(G_p)$ for every G -torsor τ over every $F \supseteq k$.*

[proved; (b) uses the inequality $u_{i+1} \geq pu_i$ for upper breaks of cyclic p^n -extensions of $k((x))$ [23, 34] and the existence results [24, 25], both cited]

Proof. (a) If F is finite, $\text{ed}_k(\tau) = 0$. Let F be infinite, $X = {}^{\tau}C$, $\infty' \in X(F)$ the twist of ∞ . The non-free locus $N \subset C$ is a finite G -stable set (fixed points of the nontrivial elements of a group acting faithfully on a curve), so ${}^{\tau}N$ is a finite set of closed points of X . The Weierstrass semigroup of ∞' on X equals $H(\infty)$ (Riemann–Roch dimensions are invariant

under base change, and $(X, \infty')_{\bar{F}} \cong (C, \infty)_{\bar{F}}$, so there is $f \in F(X)$ with pole divisor exactly $\lambda \infty'$, $\lambda = \lambda(C, \infty)$. The fibres $f = c$, $c \in F$, are pairwise disjoint effective F -rational divisors of degree λ not containing ∞' ; all but finitely many avoid ${}^{\tau}N$. For such a fibre the degrees of its closed points sum to λ , which is prime to p , so one closed point has degree e prime to p , $e \leq \lambda$, and lies in the free locus; its residue field F' is separable over F (its degree is prime to p), and Theorem 10.13 with $Z = C^{\text{free}}$ gives $\text{ed}_k(\tau \otimes F') \leq 1$. The chain of inequalities is Proposition 11.17.

(b) The upper bound is Theorem 11.3. For a connected $\mathbb{Z}/p^n$ -cover $C \rightarrow \mathbb{P}^1$ étale over $\mathbb{A}^1$ the argument in the proof of Theorem 11.3 applies verbatim ($C \rightarrow C/\langle g^{p^{n-1}} \rangle$ is an Artin–Schreier cover étale over the affine part, so the coordinate ring of $C \setminus \{\infty\}$ is free with basis $1, y, \dots, y^{p-1}$), and shows $\lambda(C, \infty) = l_n(C, \infty)$, the last lower ramification break at ∞ . Writing the upper breaks as u_1 and $u_{j+1} = pu_j + \delta_j$ with $\delta_j \geq 0$ [23, 34], Herbrand’s formula gives $l_n = u_1 + \sum_{j=1}^{n-1} p^j((p-1)u_j + \delta_j)$, which is minimal for $u_1 = 1$ and all $\delta_j = 0$, i.e. for the upper breaks $(1, p, \dots, p^{n-1})$ of $\bar{U}_n$ (Appendix B). Sharpness at $p^n = 4$ is Theorem 11.1; the last claim is Proposition 3.4 applied to an arbitrary torsor $(\text{ed}_k^{[p^{n-1}]}(\tau_a) \leq \text{trdeg}_k k(a_0) \leq 1)$.

(c) $T/G_p \rightarrow \text{Spec } F$ is étale of degree $[G : G_p]$, so it has a field factor F_1 with $p \nmid [F_1 : F] \leq [G : G_p]$; over F_1 the torsor has a point of T/G_p , hence $\tau \otimes F_1 \cong \text{Ind}_{G_p}^G \tau_p$ for a G_p -torsor τ_p over F_1 . By (a) there is F_2/F_1 with $p \nmid [F_2 : F_1] \leq \lambda_k(G_p)$ and $\text{ed}_k(\tau_p \otimes F_2) \leq 1$, and $\text{ed}_k(\tau \otimes F_2) = \text{ed}_k(\tau_p \otimes F_2)$ by Lemma 10.14. $\square$

Remark 11.20. (i) What is new relative to [32] and to Proposition 3.7(b),(c): the explicit bound, its uniformity in τ , the identification of the optimal bound from one-point covers with a local ramification invariant for cyclic groups, and the versality of the profile (Proposition 11.17), which also answers the question whether the profile of τ_{gen} depends on V . The mechanism, a free point on a curve with a fixed point, is that of [31, Lemma 9.1] and Proposition 3.7(c); what replaces largeness is Riemann–Roch on the twisted curve. (ii) A crude bound valid for every (C, ∞) is $\lambda(C, \infty) \leq$ the least integer prime to p that is $\geq 2g(C)$, since $H(\infty) \supseteq [2g(C), \infty)$; at $(2, 3)$ this gives 15 against 11. (iii) Whether $\lambda_k(\mathbb{Z}/p^n) = l_n(p)$ over *all* curves with a fixed point, not only one-point covers, is open, as is the exact value of $d_1^{(p')}(\tau_{\text{gen}})$ for $n = 2$ and p odd (between 2 and $p^2 - p + 1$) and at $(2, 3)$ (in $\{3, 5, 7, 9, 11\}$). (iv) The uniform statement is a statement about $\sup_{\tau}$: by Proposition 11.17 it is equivalent to the bound for τ_{gen} alone.

APPENDIX A. WITT-VECTOR COMPUTATIONS AT $p = 2$, $n = 3$

All formulas were re-derived from the ghost components $w_0 = x_0$, $w_1 = x_0^2 + 2x_1$, $w_2 = x_0^4 + 2x_1^2 + 4x_2$ over $\mathbb{Z}$ and reduced modulo 2 (script `witt_check.py`); they agree with (3)–(4). [computed]

Negation. $-(a_0, a_1, a_2) = (a_0, a_1 + a_0^2, a_2 + a_1^2 + a_0^2 a_1 + a_0^4)$; in particular $-x \neq x$, but $\{s - \wp x\} = \{s + \wp x\}$ as x ranges over $W_3(K)$.

The general translate. For $x = (x_0, x_1, x_2) \in W_3(K)$, $s + \wp x = (t, r_1, r_2)$ with

$$\begin{aligned} t &= s_0 + x_0^2 + x_0, \\ r_1 &= s_1 + x_0(x_0 + 1)(x_0 + s_0) + \wp(x_1), \\ r_2 &= s_2 + \Phi_2(s_0, s_1, x_0, x_1) + \wp(x_2), \end{aligned}$$

where

$$\begin{aligned} (4) \quad \Phi_2 &= s_0^3(x_0^2 + x_0) + s_0 s_1(x_0^2 + x_0) + s_0(x_0^6 + x_0^4) + s_0(x_0^2 + x_0)(x_1^2 + x_1) \\ &\quad + s_1(x_0^3 + x_0^2) + s_1(x_1^2 + x_1) + x_0^7 + x_0^4 + (x_0^3 + x_0^2)(x_1^2 + x_1) + x_1^3 + x_1^2. \end{aligned}$$

Thus the u -fibre family of Theorem 6.2 is the x_1 -independent part of r_1 : $(s + \wp(c, x_1))_1 - (c(c + 1)(c + s_0) + s_1) = \wp(x_1)$, and $(s - \wp x)_1 - (s + \wp x)_1 = x_0^4 + x_0^2 \in \wp K$.

APPENDIX B. GENUS, RAMIFICATION AND WEIERSTRASS SEMIGROUP OF $\bar{U}_n$

Ramification at ∞ . $\bar{U}_n \rightarrow \mathbb{P}_t^1$ is a $\mathbb{Z}/p^n$ -cover, étale over $\mathbb{A}^1$ and totally ramified at ∞ , of Witt class $[t]$ with $v_\infty(t) = -1$. By the conductor formula for Artin–Schreier–Witt extensions [34] (upper breaks $u_j = \max_{i < j} p^{j-1-i} m_i$ for reduced pole orders m_i) the upper breaks are $u_j = p^{j-1}$, and Herbrand’s function converts them to the lower breaks

$$l_1 = 1, \quad l_{j+1} = l_j + p^j(u_{j+1} - u_j) = l_j + (p-1)p^{2j-1},$$

which gives (1).

[cited] Direct verification at (2, 3) [computed, `genus_U3.py`]: the tower $U_3 \rightarrow U_2 = E_0 \rightarrow \mathbb{P}_{y_0}^1 \rightarrow \mathbb{P}_t^1$ has layers $y_0^2 + y_0 = t$ (break 1), $y_1^2 + y_1 = y_0^3 + y_0^2$ (break 3) and $y_2^2 + y_2 = C(y_0, y_1)$ with C from (4). On E_0 , using $y_1^2 = y_1 + y_0^3 + y_0^2$, $C = y_0^7 + y_0^6 + (y_0^3 + y_0^2)y_1$, of pole order 14; one reduction step, $C + \wp(y_0^2 y_1) = y_0^4 y_1 + y_0^3 y_1$, has odd pole order $8 + 3 = 11$. So the lower breaks are (1, 3, 11) and the upper breaks (1, 2, 4), as predicted.

Genus. With $l_0 := -1$, Riemann–Hurwitz gives $2g(\bar{U}_n) - 2 = -2p^n + \sum_{j=1}^n (p^{n-j+1} - 1)(l_j - l_{j-1})$, whence

$$g(\bar{U}_2) = \frac{1}{2}p(p-1)^2, \quad g(\bar{U}_3) = 7 \ (p=2), \quad g(\bar{U}_4) = 35 \ (p=2), \quad g(\bar{U}_3) = 78 \ (p=3).$$

Cross-check at (2, 3): the different exponent is $(u_1 + 1) + 2(u_2 + 1) + 4(u_3 + 1) = 2 + 6 + 20 = 28$ and $2g - 2 = -16 + 28 = 12$. [computed]

Weierstrass semigroup at (2, 3). The functions $y_0, y_1, y_2 + y_0^2 y_1$ are regular on U_3 with pole orders 4, 6, 11 at ∞ ; $\langle 4, 6, 11 \rangle$ has exactly the $7 = g$ gaps $\{1, 2, 3, 5, 7, 9, 13\}$, so it is the full semigroup of ∞ . The smallest odd pole order of a function regular on U_3 is therefore 11.

Degrees realised on $\bar{U}_n$. Let H be the semigroup of ∞ on $\bar{U}_n$. For $d \in H$ choose $f \in K(\tau_{\text{gen}} \bar{U}_n)$ with pole divisor $d\infty''$ and $df \neq 0$ (possible for every $d \in H$ prime to p , and at (2, 3) for every $d \in \langle 4, 6, 11 \rangle$, e.g. $y_0, y_1, y_0^2 + y_1, y_0 y_1$ for 4, 6, 8, 10). Then $K(\tau_{\text{gen}} \bar{U}_n)$ is a separable extension of degree d of $K(f)$, and since $K = k(s_0, \dots, s_{n-1})$ is Hilbertian [22], for infinitely many $c \in K$ the fibre $f = c$ is a single closed point of degree d in the free locus. Hence

$$D(U_n) \supseteq \{d \in H \setminus \{0\} : d \text{ prime to } p\}, \quad D(U_3) \supseteq \langle 4, 6, 11 \rangle \setminus \{0\}, \quad D(U_2) = \mathbb{Z}_{\geq 2} \ (p=2).$$

With Corollary 11.7, $D(U_3) \cap \{1, 2\} = \emptyset$; the unknown part of $D(U_3)$ is $\{3, 5, 7, 9, 13\}$, and the odd degrees realised on $\bar{U}_3$ so far are the odd elements ≥ 11 of $\langle 4, 6, 11 \rangle$ (13 is a gap). [proved, modulo Hilbert irreducibility]

APPENDIX C. CYCLIC 2-POWER ACTIONS ON CURVES IN CHARACTERISTIC 2

Let C be a smooth projective curve over k with a faithful action of $G = \mathbb{Z}/2^n$, $\sigma(\cdot)$ the p -rank. For a G -orbit with inertia of order 2^a and upper breaks $u_1 < \dots < u_a$ one has $u_1 \geq 1$ and $u_{i+1} \geq 2u_i$ [23, 34]; the different exponent at a point of that orbit is $\sum_{i=1}^a (2^i - 2^{i-1})(u_i + 1)$ in terms of the upper breaks of the inertia group. Deuring–Shafarevich [20]:

$$\sigma(C) - 1 = 2^n(\sigma(C/G) - 1) + \sum_{\text{orbits}} 2^n \left(1 - \frac{1}{e_{\text{orbit}}}\right).$$

Lemma C.1. *If $\sigma(C) = 0$ then $\sigma(C/G) = 0$, and $C \rightarrow C/G$ is totally ramified at exactly one point and étale elsewhere. Conversely such a cover of a p -rank-0 curve has p -rank 0. [proved]*

Proof. $2^n \sigma(C/G) + \sum 2^n(1 - 1/e) = 2^n - 1$; each orbit with $e \geq 2$ contributes at least 2^{n-1} , and $2^n - 1$ is reached only by a single orbit with $e = 2^n$. $\square$

Lemma C.2. *Let E be the supersingular elliptic curve over k . Every faithful $\mathbb{Z}/4$ -action on E is conjugate in $\text{Aut}(E)$ to one fixing O , and the six elements of order 4 of $\text{Aut}(E, O) \cong \text{SL}_2(\mathbb{F}_3)$ form one conjugacy class; so up to isomorphism of G -curves (and replacing the generator by its inverse) $(E, \mathbb{Z}/4)$ is $(E_0, \text{Witt action})$. There is no faithful $\mathbb{Z}/8$ -action on E (Lemma 2.5(iii)). [proved]*

Proof. Write $\varphi = t_P \circ \alpha$ with $\alpha \in \text{Aut}(E, O)$. If $\alpha = 1$, $\text{ord } \varphi = \text{ord } P$ is odd ($E(k)[2] = 0$); if $\alpha = -1$, $\varphi^2 = t_{P+\alpha P} = 1$. So order 4 forces $\text{ord } \alpha = 4$, and $t_Q \varphi t_{-Q} = t_{(1-\alpha)Q} \alpha$ with $1 - \alpha$ a nonzero isogeny, surjective on $E(k)$; choose Q with $(1 - \alpha)Q = -P$. The conjugacy classes of $\text{SL}_2(\mathbb{F}_3)$ are $\{1\}$, $\{-1\}$, two of order 3, two of order 6 and one of order 4, of size 6. $\square$

Enumeration (`rh_enum.py`, output `rh_enum.out`) [computed]. Imposing only the necessary conditions above (integrality and non-negativity of genus and p -rank at every level of the tower $C \rightarrow C/\langle g^{2^{n-1}} \rangle \rightarrow \cdots \rightarrow C/G$, Hasse–Arf, $u_{i+1} \geq 2u_i$), the possible $(g(C), g(C/G), \sigma(C/G), \sigma(C))$ for faithful $\mathbb{Z}/8$ -curves with $g(C) \leq 10$ are:

$g(C)$	C/G	ramification	$\sigma(C)$
1	ordinary elliptic	free action (translation by an 8-torsion point)	1
5	ordinary elliptic	one orbit with inertia $\mathbb{Z}/2$	5
7	$\mathbb{P}^1$	one totally ramified point, lower breaks (1, 3, 11)	0
9	$\mathbb{P}^1$	one totally ramified point, lower breaks (1, 3, 15)	0
9	genus 1 or 2	various (one or two orbits, or free)	5, 7, 9 or 1

Consequences: a faithful $\mathbb{Z}/8$ -curve of genus ≤ 6 is an ordinary elliptic curve with translation action or has genus 5 and p -rank 5; no faithful $\mathbb{Z}/8$ -curve has genus 2, 3, 4, 6, 8; and a p -rank-0 one of genus ≤ 10 has genus 7 or 9, with $C/G = \mathbb{P}^1$, $\mathbb{Z}/4$ -quotient a supersingular elliptic curve and $\mathbb{Z}/2$ -quotient $\mathbb{P}^1$. For $\mathbb{Z}/4$ and $g(C) \leq 6$: the p -rank-0 entries have a single totally ramified point with lower breaks $(1, 2j+1)$ (genus j) or C/G of genus 1; the only other genus-1 entry is an ordinary elliptic curve with translation by a 4-torsion point. A genus-2 faithful $\mathbb{Z}/4$ -curve has p -rank 0, quotient $\mathbb{P}^1$ and lower breaks $(1, 5)$.

APPENDIX D. THE LEVEL-2 SET $\mathfrak{F}$ AND THE ISOGENY MECHANISM

Recall $\mathfrak{F} = \{f \in K/\wp K : \text{ed}_k((s_0, s_1)_K \otimes K(v_f)) = 1\}$ (Theorem 11.6(c)). Call a descent of $(s_0, s_1)_K \otimes K(v_f)$ “of Case A” if its descent field lies inside K .

Proposition D.1 (level-2 Case A is explicit). *f admits a level-2 Case-A descent, i.e. $(s_0, s_1) \equiv \beta(t)$ over $K(v_f)$ with $t \in K$ and $\beta \in W_2(k(T))$, iff*

$$f \equiv s_1 + x_0(x_0 + 1)(x_0 + s_0) + \beta_1(t) \pmod{\wp K}$$

for some $x_0, t \in K$ and $\beta_0, \beta_1 \in k(T)$ with $\beta_0(t) = s_0 + x_0^2 + x_0$. This set $\mathfrak{F}_A$ contains $\{s_1 + x_0(x_0 + 1)(x_0 + s_0) + b(s_0 + x_0^2 + x_0) : x_0 \in K, b \in k(T)\}$ (the case $\beta_0 = T$). [proved]

Proof. “ $\Leftarrow$ ”: over $K' = K(v_f)$ put $x_1 := v_f + h$ where $f = s_1 + x_0(x_0 + 1)(x_0 + s_0) + \beta_1(t) + \wp h$; by Appendix A, $(s_0, s_1) + \wp(x_0, x_1) = (\beta_0(t), \beta_1(t))$. “ $\Rightarrow$ ”: the descent field is $k(t)$ (Theorem 11.6(a)); (s_0, s_1) and $\beta(t)$ agree over K' , so $(s_0, s_1) \equiv \beta(t) + \varepsilon(0, f)$ over K with $\varepsilon = 1$ since $\text{ed}_k((s_0, s_1)_K) = 2$; read off the coordinates of $(s_0, s_1) + \wp(x_0, x_1) = (\beta_0(t), \beta_1(t) + f)$. $\square$

Proposition D.2 (the E_0 -part of $\mathfrak{F}$). *$\{f : E'(K(v_f)) \neq \{\infty'\}\} = \{s_1 + u(u+1)(u+s_0) : u \in K\} + \wp K \subset \mathfrak{F}_A$. Equivalently, the quadratic twist $E'^{(f)} : w^2 + w = u(u+1)(u+s_0) + s_1 + f$ has an affine K -point iff f is in this family, and never one with $u \notin K$. [proved]*

Proof. As in Corollary 11.7, every $Q \in E'(K')$ has $\sigma Q = -Q$, so $u(Q) \in K$; then $w \in K' \setminus K$ with $w^2 + w \in K$ forces $w = a + v_f$, $a \in K$, and $f \equiv s_1 + u(u+1)(u+s_0)$. Conversely such f gives the point $(u, v_f + a)$. $\square$

Proposition D.3 ($f \in k(s_0)$). *For $e \in k(s_0)$ with $e \notin \{0, s_0\} + \wp k(s_0)$: $\text{ed}_k(\tau_2 \otimes K_2(v_e)) = 2$. Hence $k(s_0) \cap \mathfrak{F} = \{s_0\} + \wp K$.* [proved]

Proof. The total space of $\tau_2 \otimes K_2(v_e)$ is $k(z_0, v_e)(z_1) = M(z_1)$ with $M = k(z_0, v_e)$ fixed by $g^2: (z_0, z_1) \mapsto (z_0, z_1 + 1)$; apply Proposition 11.8 for $\mathbb{Z}/4$. (No genus hypothesis on the curve $v^2 + v = e(z_0^2 + z_0)$ is needed; for $e \in k(s_0)$ the twist $E'^{(e)}$ is isomorphic to E' via $s_1 \mapsto s_1 + e$, which shows directly $E'^{(e)}(K_2) = 0$, consistent with Proposition D.2.) $\square$

Proposition D.4 (corestriction exclusion of constant descents). *Let $M_0 \subset K$ be a subfield algebraically closed in K , $f' \in M_0$ with $f \equiv f' \pmod{\wp K}$, $M = M_0(v_{f'})$, so $K' = K \otimes_{M_0} M$. If $s_0 \notin M_0 + \wp K$ (e.g. $M_0 \subset k(s_1, s_2)$), then no descent field of $(s_0, s_1)_K \otimes K'$, in particular none of $\tau_{\text{gen}} \otimes K'$, is contained in M .* [proved]

Proof. Let $(s_0, s_1) \equiv \beta$ over K' with $\beta \in W_2(M)/\wp$. Corestriction is compatible with base change along $M_0 \rightarrow K$, so $\text{cores}_{K'/K}(\beta) = \gamma$ comes from $W_2(M_0)/\wp$, while $\text{cores}_{K'/K} \text{res}(s_0, s_1) = 2(s_0, s_1) = VF(s_0, s_1) = (0, s_0^2) \equiv (0, s_0)$. So $(0, s_0) \equiv (\gamma_0, \gamma_1)$ over K ; $\gamma_0 \in \wp K \cap M_0 = \wp M_0$, so after modifying γ by $\wp(h, 0)$, $\gamma_0 = 0$ and $s_0 + \gamma_1 \in \wp K$ by (5), contradicting $s_0 \notin M_0 + \wp K$. $\square$

This does not exclude a general Case-B descent: for a σ -stable descent field it only says that $(0, s_0) = \text{Ind}(s_0)$, which already has $\text{ed} = 1$, descends to it.

The isogeny mechanism. By Theorem 11.10(a) and Appendix C, a witness curve of genus ≤ 6 is an ordinary elliptic curve C with G acting by translation by a point P of order 8, so $C' = C/\langle P \rangle$ is ordinary elliptic, $C \rightarrow C'$ is the étale $\mathbb{Z}/8$ -isogeny (the dual of V^3 , i.e. $C = C'^{(8)}$), and C' is dominated by $D_{f,a}$ for general a : the Jacobians of the curves $D_{f,a}$ must have a *fixed* ordinary elliptic factor, which requires $f(a, s_2)$ to have at least two poles on the s_2 -line. The class of $\tau_{\text{gen}} \otimes K'$ is then $\delta(Q)$ for the connecting map $\delta: C'(K') \rightarrow H^1(K', \mathbb{Z}/8)$ of $0 \rightarrow \mathbb{Z}/8 \rightarrow C \rightarrow C' \rightarrow 0$ and the non-constant point $Q \in C'(K')$ given by $k(C') \subset K'$. For $n = 1$ and $C': y^2 + xy = x^3 + a_6$ (ordinary, $j = 1/a_6$), the substitution $y = xz$ gives $z^2 + z = x + a_6/x^2$, so the étale $\mathbb{Z}/2$ -isogeny torsor at (x, y) has Artin-Schreier class $x + a_6/x^2$; the $\mathbb{Z}/8$ version is the Witt vector of the tower $C'^{(8)} \rightarrow C'^{(4)} \rightarrow C'^{(2)} \rightarrow C'$, not computed here. Since δ is a homomorphism and s is σ -invariant, $\delta(Q + \sigma Q) = 2s \equiv (0, 0, s_0)$ and $\delta(Q - \sigma Q) = 0$, i.e. $Q - \sigma Q$ lifts to $C(K')$. We have not derived a contradiction from these constraints. [open]

APPENDIX E. THE CUBIC SLOT ON $\bar{U}_3$: WITT TRACE, CHAIN CONDITIONS, SEARCHES

Witt trace. For a cubic $u^3 + Au^2 + Bu + C$ with $A = 1 + s_0 + \alpha^2$, $B = s_0 + \alpha$, $C = s_1 + \beta^2 + \beta$ and $t = s_0 + u^2 + u$, the trace $[t] + [t'] + [t'']$ has coordinates $(e_1, e_2, e_1 e_3 + e_1^2 e_2)$ in the elementary symmetric functions of t, t', t'' (from (3); the ghost components of $\sum [t_i]$ are the power sums $\sum t_i^{2^m}$). Moreover $3x = (x_0, x_1 + x_0^2, x_2 + x_1^2 + x_0^2 x_1)$. The scripts `cores_cubic*.py` compute $d := 3s - \text{Tr}[t]$: $d_0 = \wp(A)$; after subtracting $\wp[A]$ the coordinate 1 is $\wp(\alpha^4 + \alpha^3 + \alpha s_0 + \beta + s_0^2 + s_0)$, automatically in $\wp K$ (the $n = 2$ statement of Theorem 11.1 seen through `cores`); after subtracting $\wp V[y_1]$ the last coordinate is $s_2 + R(s_0, s_1, \alpha, \beta)$ with R a polynomial of 67 terms, and it agrees with $\text{Tr}_{K(u)/K}(e_Q)$ modulo $\wp K$. Reducing squares ($x^2 \equiv x$), $R \equiv R_{\text{red}}$ with

$$\begin{aligned} R_{\text{red}} = & \alpha^{11} + s_0 \alpha^{10} + s_0 \alpha^9 + \alpha^8 \beta + s_1 \alpha^8 + s_0 \alpha^7 + \alpha^6 \beta + s_1 \alpha^6 + s_0 \alpha^6 + \alpha^5 \beta^2 + \alpha^5 \beta + \alpha^5 + s_1 \alpha^5 + s_0^2 \alpha^5 \\ & + s_0 \alpha^4 \beta^2 + \alpha^4 \beta + s_0 \alpha^4 \beta + s_0 s_1 \alpha^4 + \alpha^3 \beta + \alpha^3 + s_0 \alpha^3 + s_0^3 \alpha^3 + s_0^4 \alpha^3 + s_0^2 \alpha^2 \beta + s_0 \alpha^2 + s_0^2 s_1 \alpha^2 \\ & + s_0^3 \alpha^2 + s_0^5 \alpha^2 + \alpha \beta^2 + s_0 \alpha \beta^2 + s_0^2 \alpha \beta^2 + \alpha \beta + s_0^2 \alpha \beta + \alpha + s_1 \alpha + s_0 \alpha + s_0 s_1 \alpha + s_0^2 s_1 \alpha + s_0^4 \alpha \\ & + s_0^5 \alpha + \beta^3 + s_1 \beta^2 + s_0 \beta^2 + s_0^3 \beta^2 + \beta + s_1 \beta + s_0^3 \beta + s_0^4 \beta + s_1 + s_0 s_1 + s_0^2 s_1 + s_0^3 s_1 + s_0^4 s_1. \end{aligned}$$

The twisted G -action on W_3 acts on lines by $(\alpha, \beta) \mapsto (\alpha + 1, \alpha + \beta + 1)$, $(\alpha, \beta + 1)$ and preserves (3) (checked symbolically). [computed]

Chain conditions at $s_2 = \infty$. Let $\hat{K} = K_2((1/s_2))$ and $e = \sum_{i \leq N} c_i s_2^i \in \hat{K}$, $c_i \in K_2$. Then $e \in \wp \hat{K}$ iff $c_0 \in \wp K_2$ and, for every odd $i_0 \geq 1$, with $2^J i_0$ the last index of the chain $i_0, 2i_0, 4i_0, \dots$ carrying a nonzero coefficient,

$$(\text{Ch}_{i_0}) \quad \sum_{j=0}^J c_{i_0 2^j}^{2^{J-j}} = 0 \quad \text{in } K_2.$$

(The negative part is always $\wp$ -exact; the solution $y = \sum b_i s_2^i$ of $y^2 + y = \sum_{i>0} c_i s_2^i$ has $b_{i_0} = c_{i_0}$, $b_{2m} = c_{2m} + b_m^2$, and must be a finite sum.) In particular a top coefficient $c_N \neq 0$ with N odd excludes $e \in \wp K$. Proposition 11.15(b) uses: for $\alpha = a_0 + a_1 s_2$, $\beta = b_0 + b_1 s_2$ (plus arbitrary tails of negative order), $c_{11} = a_1^{11}$ and $c_{13} = c_{14} = c_{15} = 0$; with $a_1 = 0$, $c_3 = b_1^3$ and $c_6 = c_{12} = 0$ (`deg1_family.py`). For pole orders ≥ 2 the chain equations are nonlinear in the coefficients and Laurent tails enter; no general argument is known to us.

Searches (`f2poly.py`, `search_lines.py`, `search_targeted.py`) [computed]. For $\alpha, \beta \in \mathbb{F}_2[s_0, s_1, s_2]$, membership $s_2 + R \in \wp K$ is equivalent to membership in $\wp(k[s_0, s_1, s_2])$ and is decided exactly by the monomial-chain algorithm (top monomial must be a square; subtract $\wp$ of its root; iterate). None of the following families contains a line satisfying (3): $\deg \alpha \leq 1$, $\deg \beta \leq 2$ (16384 lines); $\alpha = 0$, $\deg \beta \leq 3$ (2^{20} lines); $\alpha \in \mathbb{F}_2[s_0, s_1]$ of degree ≤ 2 with β of s_2 -degree ≤ 2 and (s_0, s_1) -degree ≤ 1 (2^{15}); α of s_2 -degree ≤ 1 and (s_0, s_1) -degree ≤ 1 with β of s_2 -degree ≤ 2 resp. ≤ 3 (2^{15} resp. 2^{18}); α, β both of total degree ≤ 2 (2^{20}). A finite-field specialisation test (`ff_test.py`: specialise $(s_0, s_1, s_2) \in \mathbb{F}_8^3$, take the roots $u \in \mathbb{F}_{512}$ of the cubic and compute the absolute trace of $e_Q(u)$) is available as an independent check of the full condition $e_Q \in \wp K(u)$ on any candidate; it has not been needed.

APPENDIX F. SCRIPTS

All scripts are Python 3 and live in `notes/scripts/` of the repository: `witt_check.py` (ghost-component verification of (3)–(4), prints Φ_2), `wittlib.py`, `ed2_witt.py` (negation, $s \pm \wp x$), `genus_U3.py` (Artin–Schreier reduction on E_0 , $g(\bar{U}_3) = 7$), `rh_enum.py` (ramification enumeration, output `rh_enum.out`), `cores_cubic.py`, `cores_cubic2.py`, `cores_cubic3.py`, `resolvent.py` (Witt trace, R , invariance under the twisted action), `build_R.py`, `f2poly.py`, `search_lines.py`, `search_targeted.py`, `deg1_family.py`, `ff_test.py`.

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